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1 Morse Homology

Let M be a closed manifold, and let

$$f : M \rightarrow \mathbb{R}$$

be a Morse function. Let

$$\varphi : \mathbb{R} \times M \rightarrow M$$

be the flow of $-\nabla_g f$.

For $p \in \text{Crit } f$, define the unstable and stable manifolds by

$$W^u(p; -\nabla_g f) = \left\{ x \in M : \lim_{t \rightarrow -\infty} \varphi^t(x) = p \right\},$$

$$W^s(p; -\nabla_g f) = \left\{ x \in M : \lim_{t \rightarrow +\infty} \varphi^t(x) = p \right\}.$$

Theorem 1. $W^u(p)$ is an embedded space and

$$\dim W^u(p) = \text{ind}_f(p).$$

Similarly, $W^s(p)$ is an embedded space and

$$\dim W^s(p) = m - \text{ind}_f(p).$$

In local coordinates, there is a cone of stable/unstable directions. Taking the flow gives the t -coordinate with each direction.

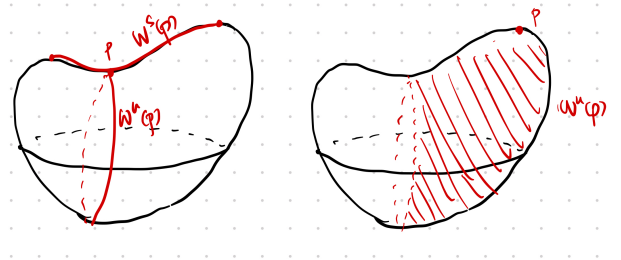


Figure 1: Stable and Unstable Manifold

Definition 1 (Morse-Smale pair). A pair (f, g) is Morse-Smale if, for every $p, q \in \text{Crit } f$,

$$W^u(p) \pitchfork W^s(q).$$

If this holds, then

$$W^u(p) \cap W^s(q)$$

is a submanifold and

$$\dim(W^u(p) \cap W^s(q)) = \text{ind}_f(p) + m - \text{ind}_f(q) - m = \text{ind}_f(p) - \text{ind}_f(q).$$

Transversality. For a fixed f , a generic metric g satisfies the Morse-Smale condition. Likewise, for fixed g , a generic f satisfies the Morse-Smale condition.

1.1 Morse Chain Complex

Define

$$MC_i(f) = \bigoplus_{\substack{p \in \text{Crit } f \\ \text{ind}(p)=i}} \mathbb{Z}\langle p \rangle.$$

For a Morse-Smale pair (f, g) , define

$$\partial_g : MC_i(f) \rightarrow MC_{i-1}(f)$$

by

$$\partial_g p = \sum_{\substack{q \in \text{Crit } f \\ \text{ind}(q)=i-1}} \langle \partial p, q \rangle q.$$

Here

$$\langle \partial p, q \rangle = \#(W^u(p) \cap W^s(q)) / \mathbb{R},$$

where the quotient is by translation along the flow line.

Theorem 2. *If (f, g) is Morse-Smale, then*

$$MH(f; g) \cong H_*(M).$$

For a self-indexing function f , set

$$C_k = H_k(M_k, M_{k-1}), \quad M_k = f^{-1}((-\infty, k]).$$

The boundary map is the connecting homomorphism

$$\delta : C_k \rightarrow C_{k-1},$$

so that

$$H_{k+1}(M_k, M_{k-1}) \rightarrow H_{k+1}(M_{k+1}, M_{k-1}) \rightarrow H_{k+1}(M_{k+1}, M_k) \xrightarrow{\delta} H_k(M_k, M_{k-1}) \rightarrow \cdots.$$

The Morse chain complex is identified with the cellular complex:

$$(MC(f), \partial) \cong (C, \delta).$$

Claim 1. There are isomorphisms

$$H_k(M_k, M_{k-1}) \cong H_k(M_k, M_{k-\frac{1}{2}}) \cong \bigoplus_{\substack{p \in \text{Crit } f \\ \text{ind } p = k}} \mathbb{Z}\langle D^u(p) \rangle.$$

Under the correspondence

$$D^u(p) := W^u(p) \cap \{f \geq k - 1/2\} \longleftrightarrow p,$$

we obtain

$$C_k \cong MC_k(f).$$

Before proving $\partial \cong \delta$, first prove the homology statement.

Claim 2.

$$H_k(C, \delta) \cong H_k(M).$$

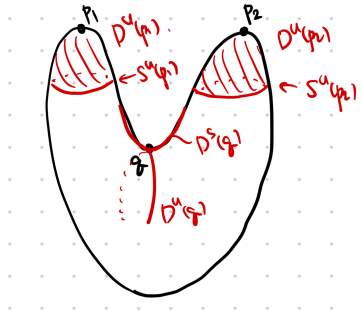


Figure 2: Boundary Operator proof

Using cellular homology, H_k is determined on the $(k+1)$ -skeleton: by the way k -cells and $(k+1)$ -cells are attached.

Boundary operator. Let

$$A^a, B^b \subset X^{a+b}$$

be closed manifolds. Then Thom isomorphisms

$$H_a(A) \cong H_b(X, X - A),$$

$$H_b(B) \rightarrow H_b(X) \rightarrow H_b(X, X - A).$$

That sends $[B]$ to $(A \cdot B)\tau$.

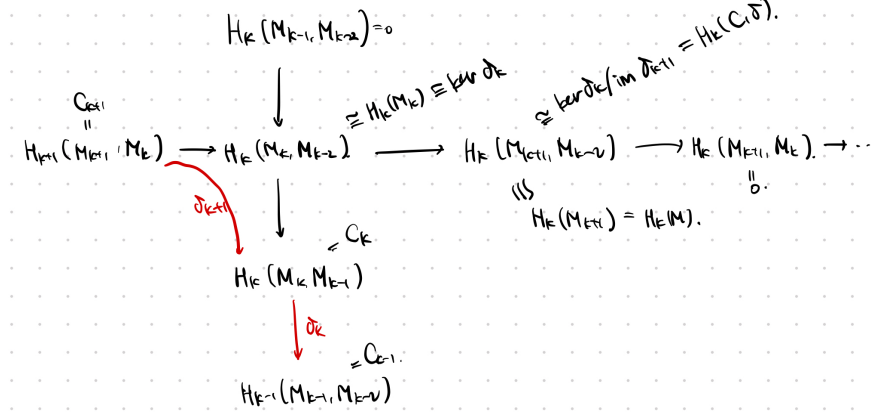


Figure 3: Boundary Operator proof

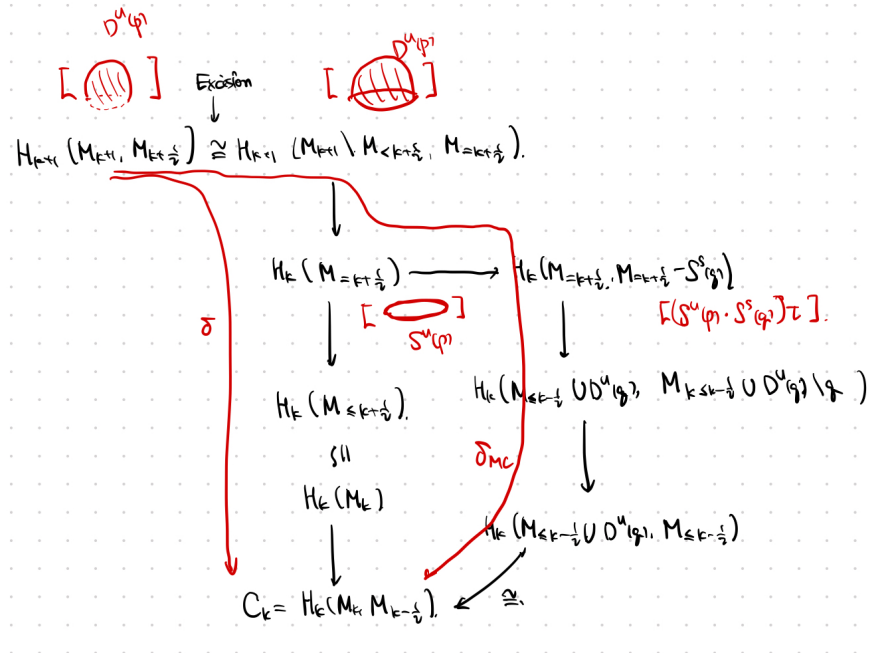


Figure 4: Boundary Operator proof

2 Morse Homology: Modern Viewpoint

For $p, q \in \text{Crit } f$, define

$$\widehat{\mathcal{M}}(p, q) = \left\{ u : \mathbb{R} \xrightarrow{C^\infty} M \mid \partial_s u = -\nabla_g f(u), \lim_{s \rightarrow -\infty} u(s) = p, \lim_{s \rightarrow +\infty} u(s) = q \right\}.$$

If $p \neq q$, there is an $\mathbb{R}$ -action by translation,

$$\mathbb{R} \times \widehat{\mathcal{M}}(p, q) \rightarrow \widehat{\mathcal{M}}(p, q), \quad (a, \gamma) \mapsto a^* \gamma = \gamma(\cdot + a).$$

If (f, g) is Morse-Smale, then

$$\widehat{\mathcal{M}}(p, q) \cong W^u(p) \cap W^s(q)$$

has a smooth manifold structure, and

$$\dim \widehat{\mathcal{M}}(p, q) = \text{ind}_f(p) - \text{ind}_f(q).$$

The quotient

$$\mathcal{M}(p, q) = \widehat{\mathcal{M}}(p, q)/\mathbb{R}$$

is a manifold of dimension

$$\text{ind}_f(p) - \text{ind}_f(q) - 1.$$

Thus

$$\begin{aligned} MC_i(f) &= \bigoplus_{\substack{p \in \text{Crit } f \\ \text{ind } p = i}} \mathbb{Z}\langle p \rangle, & \partial_g : MC_i(f) &\rightarrow MC_{i-1}(f), \\ p &\longmapsto \sum_q \# \mathcal{M}(p, q) q. \end{aligned}$$

Morse homology can be viewed as counting the number of solutions of the flow equation. This viewpoint has more applications.

2.1 Floer Homology: Hamiltonian Case

Let $(\mathbb{R}^{2n}, i)$ be a complex vector space and let

$$H : \mathbb{R}^{2n} \rightarrow \mathbb{R}$$

be a Hamiltonian. The Hamiltonian vector field is

$$X_H = -i \text{ grad } H.$$

The goal is to find a 1-periodic orbit of X_H :

$$\dot{\gamma}(t) = X_H(\gamma(t)), \quad \gamma : S^1 \rightarrow \mathbb{R}^{2n}.$$

Let

$$\mathcal{A}_H : C^\infty(\mathbb{R}/\mathbb{Z}, \mathbb{R}^{2n}) \rightarrow \mathbb{R}^{2n}$$

be the action functional

$$\gamma \longmapsto \int_0^1 \frac{1}{2} i \dot{\gamma}(t) \cdot \gamma(t) dt - \int_0^1 H(\gamma(t)) dt.$$

Then

$$\gamma \in \text{Crit } \mathcal{A}_H \iff \dot{\gamma}(t) = X_H(\gamma(t)).$$

Indeed,

$$\begin{aligned} (d\mathcal{A}_H(\gamma))(Y) &= \int_0^1 \frac{1}{2} \{i\dot{Y} \cdot \gamma + i\dot{\gamma} \cdot Y\} dt - \int_0^1 dH(\gamma)Y dt \\ &= \int_0^1 i\dot{\gamma} \cdot Y dt - \int_0^1 X_H(\gamma) \cdot Y dt \\ &= \int_0^1 i(\dot{\gamma} - X_H(\gamma))Y dt. \end{aligned}$$

However, defining the manifold structure is hard. Even if it is defined successfully, the spaces W^u and W^s are problematic. The moduli space $\mathcal{M}(x, y)$ should be understood in Floer's sense. Use the L^2 -metric: for $u_1, u_2 \in \gamma^* T\mathbb{R}^{2n}$,

$$\langle u_1, u_2 \rangle_{L^2} = \int_0^1 u_1 \cdot u_2 dt.$$

With the L^2 metric,

$$d\mathcal{A}_H(\gamma)Y = \langle \nabla_{L^2} \mathcal{A}_H(\gamma), Y \rangle_{L^2} = \int_0^1 i(\dot{\gamma} - X_H(\gamma))Y dt,$$

so

$$\nabla_{L^2} \mathcal{A}_H(\gamma) = i(\dot{\gamma} - X_H(\gamma)).$$

A trajectory $u : \mathbb{R} \rightarrow C^\infty(\mathbb{R}/\mathbb{Z}, \mathbb{R}^{2n})$ is equivalent to

$$u : \mathbb{R} \times \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{R}^{2n},$$

and the negative gradient equation is

$$\partial_s u + \nabla_{L^2} \mathcal{A}_H(u) = 0, \quad \text{i.e.} \quad \partial_s u + i\partial_t u - X_H(u) = 0.$$

The first equation is an ODE in an infinite-dimensional space, while the second is a PDE in a finite-dimensional space.

For critical points $x, y \in \text{Crit } \mathcal{A}_H$, set

$$\widehat{\mathcal{M}}(x, y) = \left\{ u \mid \partial_s u + i\partial_t u - X_H(u) = 0, \lim_{s \rightarrow -\infty} u(s, \cdot) = x, \lim_{s \rightarrow +\infty} u(s, \cdot) = y \right\}.$$

For a nice Hamiltonian H , this is a smooth manifold of dimension

$$\text{ind}(x) - \text{ind}(y).$$

The Floer chain complex is

$$FC_i(H) = \bigoplus_{\substack{\alpha \in \text{Crit } \mathcal{A}_H \\ \text{ind}(\alpha) = i}} \mathbb{Z}\langle \alpha \rangle, \quad \partial_i : FC_i(H) \rightarrow FC_{i-1}(H),$$

$$x \mapsto \sum_y \# \mathcal{M}(x, y) y.$$

One has

$$\partial \circ \partial = 0,$$

and the Floer homology is well-defined.

2.2 Lagrangian Floer Homology

Let L be a closed submanifold of either $(\mathbb{R}^{2n}, \sum_i dx_i \wedge dy_i)$ or (M^{2n}, ω) , of dimension n , such that

$$\omega|_L = 0.$$

Then L is Lagrangian.

Examples include $\mathbb{R}^n \times \{0\} \subset \mathbb{R}^{2n}$, the zero section of T^*X , simple closed curves on a genus- g Riemannian surface, and $\mathbb{RP}^n \subset \mathbb{CP}^n$.

Let L_0, L_1 be Lagrangians and assume

$$\frac{1}{2} \sum_i (x_i dy_i - y_i dx_i)|_{L_i} = df_i, \quad f_i \in C^\infty(L_i).$$

Define

$$\mathcal{P}(L_0, L_1) = \{\gamma : [0, 1] \rightarrow \mathbb{R}^{2n} : \gamma(0) \in L_0, \gamma(1) \in L_1\}.$$

The action functional is

$$A(\gamma) = \int_0^1 \frac{1}{2} i\dot{\gamma}(t) \cdot \gamma(t) dt - f_0(\gamma(0)) + f_1(\gamma(1)).$$

Then

$$dA(\gamma)Y = \int_0^1 i\dot{\gamma}Y dt,$$

and

$$\gamma \in \text{Crit } A \iff \gamma \text{ is constant} \iff \gamma \in L_0 \cap L_1.$$

For $u : \mathbb{R} \rightarrow \mathcal{P}(L_0, L_1)$,

$$\partial_s u + \nabla_{L^2} A(u) = 0 \iff \partial_s u + i\partial_t u = 0,$$

so u is holomorphic. The chain group is

$$FC(L_0, L_1) = \bigoplus_{p \in L_0 \cap L_1} \mathbb{Z}\langle p \rangle.$$

3 Fredholm Operators and Banach Spaces

Definition 2 (Fredholm operator). *Let A, B be Banach spaces and let*

$$L : A \rightarrow B$$

be a bounded linear operator. Then L is Fredholm if

$$\dim \ker L < \infty, \quad \dim \operatorname{coker} L < \infty.$$

The Fredholm index is

$$\operatorname{ind} L = \dim \ker L - \dim \operatorname{coker} L.$$

Remark 1. *A bounded linear operator $L : A \rightarrow B$ is Fredholm iff there exists a bounded linear operator $D : B \rightarrow A$ such that*

$$DL - I_A, \quad LD - I_B$$

are compact.

3.1 Compactness (Reference: Stein and Shakarchi)

A subset X of a Hilbert space $\mathcal{H}$ is compact if every sequence in X has a subsequence converging in norm to an element of X .

An operator $T : \mathcal{H} \rightarrow \mathcal{H}$ is compact if

$$T(B) = \{g \in \mathcal{H} : g = T(f) \text{ for some } f \in B\}$$

has compact closure, where

$$B = \{f \in \mathcal{H} : \|f\| \leq 1\}.$$

Proposition 1. *Let S be compact on $\mathcal{H}$ and let T be a bounded linear operator. Then ST and TS are compact. If T_n are compact operators and*

$$\|T_n - T\| \rightarrow 0,$$

then T is compact. Also, T is compact iff it is the norm limit of finite-rank operators. Finally,

$$T \text{ compact} \iff T^* \text{ compact}.$$

Sketch. For example, choose an orthonormal basis $\{e_i\}_{i=1}^\infty$ of $\mathcal{H}$ and let Q_n be the orthogonal projection onto $\operatorname{span}\{e_i : i \geq n\}$. Then

$$\|Q_n T\| \rightarrow 0,$$

and if $I = P_n + Q_n$, then

$$\|T - P_n T\| \rightarrow 0.$$

□

Remark 2. *If T can be diagonalized, then*

$$T \text{ compact} \iff |\lambda_n| \rightarrow 0.$$

Remark 3. *Every Hilbert-Schmidt operator is compact.*

Theorem 3 (Spectral theorem). *Let T be a compact symmetric operator on a Hilbert space $\mathcal{H}$. Then there exists an orthonormal basis of $\mathcal{H}$ consisting of eigenvectors of T , say*

$$\{\varphi_k\}_{k=1}^\infty, \quad \lambda_k \in \mathbb{R}, \quad |\lambda_k| \rightarrow 0.$$

Properties. If $K \subset A$ is finite-dimensional, then there exists a closed subspace $A_0 \subset A$ such that

$$A \cong K \oplus A_0.$$

Indeed, if $\{e_1, \dots, e_n\}$ is a basis of K , the Hahn-Banach theorem gives $e_1^*, \dots, e_n^* \in A^*$ and

$$A_0 = \bigcap_{i=1}^n \ker e_i^*.$$

3.2 Basic Results on Banach Spaces (Reference: Stein and Shakarchi)

Theorem 4 (Hahn-Banach). *Let $V_0 \subset V$ be a linear subspace. If ℓ_0 is a bounded linear functional on V_0 and*

$$\ell_0(v) \leq p(v), \quad v \in V_0,$$

where p is sublinear, then there exists ℓ on V extending ℓ_0 such that

$$\ell(v) \leq p(v), \quad v \in V.$$

Corollary 1. *If $f_0 \in B^*$ and $\|f_0\| = M$, then there exists $\ell \in B^*$ such that*

$$\ell(f_0) = M, \quad \|\ell\|_{B^*} = 1.$$

Corollary 2. *Let B_1, B_2 be Banach spaces and let $S \subset B_1$ be a dense linear subspace. If*

$$T_0 : S \rightarrow B_2, \quad \|T_0(f)\|_{B_2} \leq M\|f\|_{B_1},$$

then T_0 extends to a bounded linear map $T : B_1 \rightarrow B_2$ with

$$\|T(f)\|_{B_2} \leq M\|f\|_{B_1}.$$

Theorem 5 (Baire category). *A subset $E \subset X$ is of first category if it is a countable union of nowhere dense sets. A subset $E \subset X$ is generic if E^c is of first category.*

Theorem 6 (Open mapping theorem). *Let X, Y be Banach spaces and let $T : X \rightarrow Y$ be a bounded linear transformation. If T is surjective, then T is an open mapping.*

Application. Let $B_0 = \text{im } L \subset B$. If $\dim \text{coker } L < \infty$, then there is a closed complement C such that

$$B_0 \oplus C \cong B.$$

The induced map

$$\bar{L} : A/\ker L \oplus C \rightarrow B_0 \oplus C, \quad ([a], c) \mapsto (La, c)$$

is a bounded linear bijection. By the open mapping theorem it is an isomorphism, and therefore

$$B_0 = \bar{L}(A/\ker L \oplus 0)$$

is closed.

If $P : A \rightarrow B$ is bounded linear with $\|P\| < \varepsilon$, then $L + P$ is Fredholm and

$$\text{ind}(L + P) = \text{ind}(L).$$

Write

$$L = \begin{pmatrix} L_0 & 0 \\ 0 & 0 \end{pmatrix}, \quad P = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : A_0 \oplus K \rightarrow B_0 \oplus C.$$

□

The inverse of $L_0 + a$ is given by the Neumann series

$$(L_0 + a)^{-1} = (I + L_0^{-1}a)^{-1}L_0^{-1} = [I - L_0^{-1}a + (L_0^{-1}a)^2 - \dots]L_0^{-1}.$$

If

$$\varepsilon = \frac{1}{\|L_0^{-1}\|}, \quad \|a\| \leq \|P\| < \varepsilon,$$

then

$$\|L_0^{-1}a\| < 1,$$

so $(L_0 + a)^{-1}$ exists. A block calculation gives

$$\begin{pmatrix} I & 0 \\ -c(L_0 + a)^{-1} & I \end{pmatrix} \begin{pmatrix} L_0 + a & b \\ c & d \end{pmatrix} \begin{pmatrix} I & -(L_0 + a)^{-1}b \\ 0 & I \end{pmatrix} = \begin{pmatrix} L_0 + a & 0 \\ 0 & d - c(L_0 + a)^{-1}b \end{pmatrix}.$$

Therefore

$$\dim \ker(L + P) = \dim \ker(d : K \rightarrow C),$$

$$\dim \text{coker}(L + P) = \dim \text{coker } d,$$

which are finite because K and C are finite-dimensional. Hence this works because L is Fredholm.

Definition 3 (Fredholm map). *If M, N are Banach manifolds, then $f : M \rightarrow N$ is Fredholm if*

$$d_p f : T_p M \rightarrow T_{f(p)} N$$

is Fredholm for every $p \in M$.

Theorem 7 (Sard-Smale). *If $f : M \rightarrow N$ is a C^∞ Fredholm map, then the set of regular values of f is a Baire set.*

Theorem 8 (Parametric transversality). *Let*

$$E \xrightarrow{\pi} M \times S$$

be a Banach vector bundle, and assume F has a C^∞ section. Suppose

$$(m, s) \in F^{-1}(0_E)$$

- (1) $T_{(m,s)}(M \times S) \rightarrow T_{((m,s),0)} E \xrightarrow{\pi} E_{(m,s)}$ *is surjective, and*
(2) $D_m F_s := \pi \circ d_{(m,s)} F : T_m M \rightarrow E_{(m,s)}$ *is Fredholm.*

Then for generic $s \in S$, the map $D_m F_s$ is surjective and

$$F_s^{-1}(0_E)$$

is a smooth manifold with

$$T_m(F_s^{-1}(0_E)) = \ker D_m F_s, \quad \dim F_s^{-1}(0_E) = \text{ind } D_m F_s.$$

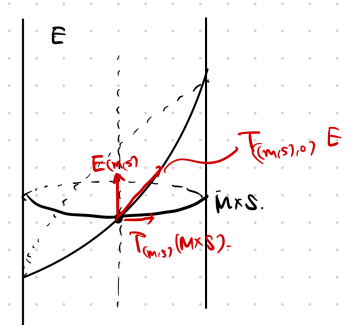


Figure 5: Transversality setting

Theorem 9. *The set of Morse functions in $C^\infty(M)$ is dense.*

Idea. For a generic $f \in C^\infty(M)$, consider the section

$$F : M \rightarrow T^*M, \quad F = df.$$

Then

$$F \pitchfork 0_{T^*M}.$$

Thus, for every $x \in F^{-1}(0_{T^*M}) = \text{Crit } f$,

$$T_x M \xrightarrow{dF} T_{(x,0)}(T^*M) \xrightarrow{\Pi} T_x^*M$$

is surjective. Since

$$\Pi \circ d_x F = \text{Hess}_x f,$$

this says that the Hessian is nondegenerate.

Equivalently, consider the Banach vector bundle

$$\begin{array}{ccc} T^*M \times C^\ell(M) & \longrightarrow & T^*M \\ \downarrow & & \downarrow \\ TM \times C^\ell(M) & \longrightarrow & M \end{array}$$

with

$$F(x, f) = (df_x, f).$$

Then

$$(x, f) \in F^{-1}(0_{T^*M} \times C^\ell(M)).$$

The projection of the differential

$$\Pi \circ d_{(x,f)} F : T_x M \times T_f C^\ell(M) \rightarrow T_x^* M, \quad (v, g) \mapsto d_x g + \text{Hess}_x f(v)$$

is surjective, because g is arbitrary. Moreover, $\Pi \circ d_x F_f : T_x M \rightarrow T_x M^*$ is finite dimensional so Fredholm. Therefore

$$C_{\text{reg}}^\ell = \{f : df \pitchfork 0_{T^*M}\}$$

is a Baire set. Since $C^\ell(M)$ is a complete metric space, C_{reg}^ℓ is dense in $C^\ell(M)$. Using the Whitney topology on $C^\infty(M)$,

$$\|f - g\|_{C^\infty} = \max_k \frac{1}{2^k} \frac{\|f - g\|_{C^k}}{1 + \|f - g\|_{C^k}},$$

we have $f_n \rightarrow f$ in C^∞ iff

$$\|f_n - f\|_{C^{\ell(n)}} \rightarrow 0$$

for some sequence $\ell(n) \rightarrow \infty$. A basis is given by open sets U such that

$$U \cap C^k \subset C^k$$

is C^k -open for all $k \in \mathbb{N}$.

Basic Morse knowledge: C_{reg}^ℓ is C^ℓ -open. Since $C^\infty \subset C^\ell$ is dense, for $f \in C^\infty$ choose $f_k \in C_{\text{reg}}^\ell$ with

$$\|f - f_k\|_{C^\ell} < \varepsilon/2.$$

Then choose $f'_k \in C^\infty$ with

$$\|f_k - f'_k\|_{C^\ell} < \varepsilon/2,$$

and f'_k is still in C_{reg}^ℓ . Hence $C_{\text{reg}}^\infty \subset C^\infty$ is C^ℓ -dense. Therefore $C_{\text{reg}}^\infty \subset C^\infty$ is dense. □

4 Banach Spaces and Banach Manifolds

The open mapping theorem implies the closed graph theorem:

$$A : E \rightarrow F \text{ is bounded} \iff \Gamma_A \subset E \oplus F \text{ is a closed subspace.}$$

A fundamental isomorphism theorem says that if

$$A \in \mathcal{L}(E, F) \text{ is surjective,}$$

then $E/\ker A$ and F are isomorphic as Banach spaces.

Theorem 10 (Uniform boundedness principle). *Let E, F be normed vector spaces with E complete, and let*

$$\{A_i\}_{i \in I} \subset \mathcal{L}(E, F).$$

If for every $e \in E$ the set $\{\|A_i e\|\}_{i \in I}$ is bounded in F , then

$$\{\|A_i\|\}_{i \in I}$$

is bounded.

Indeed, for

$$\varphi(e) = \sup_{i \in I} \|A_i e\|, \quad S_n = \{e \in E : \varphi(e) \leq n\},$$

the sets S_n are closed and $\bigcup_n S_n = E$. By Baire category, some S_n has nonempty interior. Then there exist $r > 0$ and $e_0 \in E$ such that

$$e \in B_r(e_0) \Rightarrow \varphi(e) \leq M.$$

For $\|e\| = 1$,

$$\|A_i e\| = \frac{1}{r} \|A_i(re + e_0) - A_i e_0\| \leq \frac{1}{r} \|A_i(re + e_0)\| + \frac{1}{r} \|A_i e_0\| \leq \frac{1}{r} (M + \varphi(e_0)).$$

Corollary 3. *If $A_n \in \mathcal{L}(E, F)$ converge strongly, i.e.*

$$A_n e \rightarrow A e \quad \text{for all } e \in E,$$

then $A \in \mathcal{L}(E, F)$.

4.1 Derivative

Let E, F be normed vector spaces. A map

$$f_g : U \subset E \rightarrow F$$

is tangent to another map at $u_0 \in U$ if

$$\lim_{u \rightarrow u_0} \frac{\|f(u) - g(u)\|}{\|u - u_0\|} = 0.$$

For $f : U \subset E \rightarrow F$ and $u_0 \in U$, if there is $L \in \mathcal{L}(E, F)$ such that

$$g_L(u) = f(u_0) + L(u - u_0)$$

is tangent to f at u_0 , then f is differentiable at u_0 and

$$Df : U \rightarrow \mathcal{L}(E, F).$$

Inductively,

$$D^r f = D(D^{r-1} f) : U \rightarrow L^r(E, F),$$

and if $D^r f$ is symmetric, then it lies in $L_{\text{sym}}^r(E, F)$.

Theorem 11 (Euler). *If $f : U \subset E \rightarrow F$ is C^r , then $D^r f$ is symmetric.*

Theorem 12 (Taylor). *A map $f : U \subset E \rightarrow F$ is C^r iff there are continuous maps*

$$\varphi_p : U \rightarrow L_{\text{sym}}^p(E, F), \quad p = 1, \dots, r,$$

and a remainder $R : U \rightarrow L_{\text{sym}}^r(E, F)$ such that

$$f(u+h) = f(u) + \frac{\varphi_1(u)}{1!}h + \frac{\varphi_2(u)}{2!}h^2 + \dots + \frac{\varphi_r(u)}{r!}h^r + R(u, h)h^r,$$

where

$$\varphi_p = D^p f, \quad R(u, h) = \int_0^1 \frac{(1-t)^{r-1}}{(r-1)!} (D^r f(u+th) - D^r f(u)) dt.$$

Theorem 13 (Inverse mapping theorem). *Let $f : U \subset E \rightarrow F$ be C^r , $r \geq 1$, and $u_0 \in U$. If $Df(u_0)$ is a linear isomorphism, then f is a C^r diffeomorphism from a neighborhood of u_0 to a neighborhood of $f(u_0)$. Moreover,*

$$Df^{-1}(y) = [Df(f^{-1}(y))]^{-1}.$$

Sketch. For a complete metric space M , a strong contraction $F : M \rightarrow M$ has a fixed point. Also, $GL(E, F)$ is open in $\mathcal{L}(E, F)$. Indeed, if

$$\|\psi - \varphi\| < \|\varphi^{-1}\|^{-1},$$

then

$$\|\varphi^{-1} \circ (\varphi - \psi)\| < 1,$$

so $\varphi^{-1} \circ \psi$ is invertible.

The inversion map

$$I : GL(E, F) \rightarrow GL(F, E), \quad \varphi \mapsto \varphi^{-1}$$

is C^∞ , with

$$DI(\varphi)\psi = -\varphi^{-1} \circ \psi \circ \varphi^{-1}.$$

Assume $u_0 = 0 = f(u_0)$, $E = F$, and $Df(0) = \text{id}$. Put

$$h = Df(u_0)^{-1}(f(x+u_0) - f(u_0)).$$

If

$$g(x) = x - f(x),$$

then $Dg(0) = 0$. Choose $r > 0$ such that

$$\|x\| < r \Rightarrow \|Dg(x)\| \leq \frac{1}{2}.$$

Then

$$\|g(x)\| \leq \frac{r}{2}$$

for $\|x\| < r$. For $y \in B_{r/2}(0)$, define

$$g_y(x) = y + g(x).$$

If $x_1, x_2 \in B_r(0)$, then

$$\|g_y(x_1) - g_y(x_2)\| \leq \frac{1}{2}\|x_1 - x_2\|.$$

Therefore g_y has a unique fixed point on $B_r(0)$. Hence

$$y = f(x).$$

Thus

$$f^{-1} : V_0 = D_{r/2}(0) \rightarrow U_0 = f^{-1}(D_{r/2}(0))$$

is continuous and

$$\|f^{-1}(y_1) - f^{-1}(y_2)\| \leq 2\|y_1 - y_2\|.$$

□

Theorem 14 (Implicit function theorem). *Let $U \subset E$ and $V \subset F$ be open, and let*

$$f : U \times V \rightarrow G$$

be C^r , $r \geq 1$. Suppose $(x_0, y_0) \in U \times V$ and

$$D_2 f(x_0, y_0) : F \rightarrow G$$

is an isomorphism. Then there are neighborhoods U_0 of x_0 and W_0 of $f(x_0, y_0)$ and a C^r map

$$g : U_0 \times W_0 \rightarrow V$$

such that

$$f(x, g(x, w)) = w.$$

4.2 Local Surjectivity and Local Injectivity

Theorem 15 (Local surjectivity). *Let $f : U \subset E \rightarrow F$ be C^1 . If $Df(u_0)$ is onto for some $u_0 \in U$, then f is locally onto: there are neighborhoods $U_0 \subset U$ of u_0 and $V_0 \subset F$ of $f(u_0)$ such that*

$$f|_{U_0} : U_0 \rightarrow V_0$$

is onto.

Choose a right inverse

$$T : E_0 \rightarrow F$$

using the Banach isomorphism theorem. Then

$$T^{-1} : F \rightarrow E_0$$

exists. For

$$x = u_0 + h, \quad y = f(x),$$

we solve

$$T^{-1}(y - f(u_0 + h)) = 0.$$

Using contraction mapping, one gets a sequence

$$h_n = L_{n-1} + T^{-1}\{y - f(u_0 + h_{n-1})\}$$

which is Cauchy and converges to h . Thus L is convergent and a solution exists.

Theorem 16 (Local injectivity). *Let $f : U \subset E \rightarrow F$ be C^1 . Suppose $Df(u)(E)$ is closed in F , and*

$$Df(u_0) \in GL(E, Df(u_0)(E)).$$

Then there exists a neighborhood V of u_0 such that f is injective on V , and

$$f^{-1} : f(V) \rightarrow V$$

is Lipschitz continuous.

Corollary 4. *If $U \subset E$ is open and $f : U \rightarrow F$ is C^r , $r \geq 1$, and if $Df(u_0)$ is surjective with complemented kernel, then $f(U)$ contains a neighborhood of $f(u_0)$.*

Example. Let

$$I(f) = \frac{df}{dx} + f^3.$$

Set

$$E = C^1[0, 1], \quad \|f\|_E = \sup_{x \in [0, 1]} |f(x)| + \sup_{x \in [0, 1]} \left| \frac{df}{dx}(x) \right|,$$

and

$$F = C^0[0, 1], \quad \|f\|_F = \sup_{x \in [0, 1]} |f(x)|.$$

Then

$$DI(0) = \frac{d}{dx} : E \rightarrow F$$

is surjective, since this is the fundamental theorem of calculus, and

$$\ker DI(0) = \{\text{constant functions}\}.$$

Thus, if $\|g(x)\| < \varepsilon$ on $[0, 1]$, there exists $f : [0, 1] \rightarrow \mathbb{R}$ such that

$$\frac{df}{dx} + f^3 = g.$$

4.3 Local Immersion and Submersion Theorems

Theorem 17 (Local immersion theorem). *Let $f : U \subset E \rightarrow F$ be C^r , $r \geq 1$, and let $u_0 \in U$. Suppose $Df(u_0)$ is one-to-one and its image is a closed split subspace F_1 with complement F_2 . Then there are open sets $U' \subset F_1$ and $V \subset F_1 \oplus F_2$ such that $f(u_0) \in U'$ and there is a C^r diffeomorphism*

$$\varphi : U' \rightarrow V$$

with

$$\varphi \circ f(e) = (e, 0), \quad e \in V \cap (E \times \{0\}).$$

Theorem 18 (Local submersion theorem). *Let $f : U \subset E \rightarrow F$ be C^r , $r \geq 1$, and suppose $Df(u_0)$ is surjective with split kernel E_2 and closed complement E_1 . Then there are open sets*

$$u_0 \in U' \subset U, \quad V \subset F \oplus E_2,$$

and a C^r diffeomorphism

$$\psi : V \rightarrow U'$$

such that

$$(f \circ \psi)(u, v) = u, \quad u, v \in V.$$

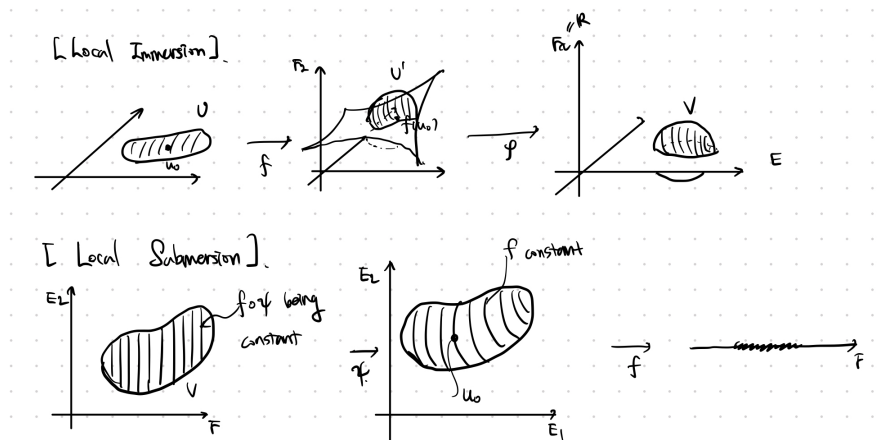


Figure 6: Local Immersion and Local Submersion

Definition 4 (Differentiable manifold). *A differentiable manifold M is a set with an atlas (U_i, φ_i) such that the charts take values in a Banach space E . Then E is called the model space. The topology is given by the charts.*

Example: Grassmann manifolds. Let E be a Banach space. The Grassmannian

$$\text{Gr}(E)$$

is the space of split subspaces of E . If $F \in \text{Gr}(E)$ and $E = F \oplus G$, set

$$U_G = \{H \in \text{Gr}(E) : E = H \oplus G\}.$$

Then U_G is a C^∞ Banach manifold modeled on $\mathcal{L}(F, G)$, with chart

$$\varphi_{F,G} : U_G \rightarrow \mathcal{L}(F, G), \quad H \mapsto \pi_G|_H \circ (\pi_F|_H)^{-1}.$$

The tangent space at $m \in M$ is

$$T_m M = \{[c]_m : c \text{ is a curve at } m\}.$$

If M is a C^{r+1} manifold, then TM is a C^r manifold. If M, N are C^{r+1} manifolds and $f : M \rightarrow N$ is C^{r+1} , then

$$Tf : TM \rightarrow TN$$

is a C^r map.

A vector bundle has local form

$$U \times F \rightarrow U,$$

where U is a base space and F is the fiber. For $u \in U$, the set $\{u\} \times F$ is called the fiber at u .

For the Grassmann bundle,

$$\rho : \text{Gr}_n(E) \rightarrow \text{Gr}(E),$$

with

$$\rho^{-1}(V) = \{F \subset E : F \text{ is an } n\text{-dimensional subspace}\}.$$

Theorem 19 (Local diffeomorphism theorem). *Let M, N be manifolds and let $f : M \rightarrow N$ be C^r , $r \geq 1$. If*

$$Tf|_{\text{fiber at } m} : T_m M \rightarrow T_{f(m)} N$$

is an isomorphism, then f is a C^r diffeomorphism from a neighborhood of m to a neighborhood of $f(m)$.

Theorem 20 (Local onto theorem). *Let $f : M \rightarrow N$ be C^r , $r \geq 1$. Suppose*

$$Tf|_{\text{fiber at } m} : T_m M \rightarrow T_{f(m)} N$$

is surjective. Then f is locally onto at m . If the kernel of $T_m f$ is split in $T_m M$, then there exist charts in which f looks like a projection.

Theorem 21 (Submersion theorem). *If $f : M \rightarrow N$ is C^∞ , n is a regular value, and $T_m f$ is surjective with split kernel, then*

$$f^{-1}(n)$$

is a closed submanifold of M , and the tangent space is

$$T_m(f^{-1}(n)) = \ker T_m f.$$

Definition 5 (Embedding). *An embedding is an injective immersion which is open or closed onto its image.*

Definition 6 (Transversality). *A C^r map $f : M \rightarrow N$ is transverse to a submanifold $P \subset N$ if $f^{-1}(P) = \emptyset$ or, for every $m \in f^{-1}(P)$,*

1. $(T_m f)(T_m M) + T_{f(m)} P = T_{f(m)} N$,
2. $(T_m f)^{-1}(T_{f(m)} P)$ splits in $T_m M$.

5 The Sobolev Model Space

Proposition 2 (Bounded embedding). *There is a bounded embedding*

$$W^{1,2}(\mathbb{R}) \hookrightarrow C^0(\mathbb{R}).$$

Proof. If $(u_n) \subset W^{1,2}$ and $u_n \rightarrow u$ in $W^{1,2}$, then after passing to a subsequence

$$u_n \in W^{1,2} \cap C^\infty.$$

Claim 1. The sequence (u_n) is equicontinuous. If

$$\|u_n\|_{W^{1,2}} \leq C,$$

then

$$|u_n(s_1) - u_n(s_0)| \leq \int_{s_0}^{s_1} |ds u_n| ds \leq |s_1 - s_0|^{1/2} \|u_n\|_{W^{1,2}} \leq C |s_1 - s_0|^{1/2}.$$

Claim 2. The sequence (u_n) is uniformly bounded. If $u_n(s_n) \rightarrow \infty$, then

$$\min u_n[s_n - 1, s_n + 1] \rightarrow \infty,$$

which contradicts boundedness in L^2 .

By Arzela-Ascoli, for each compact $K \subset \mathbb{R}$ there is a subsequence such that

$$u_n|_K \rightarrow v \in C^0(K).$$

Therefore $v = u|_K$, and u is continuous on every compact set. Hence $u \in C^0(\mathbb{R})$. □

Now let M be a smooth manifold embedded in $\mathbb{R}^N$. Define

$$W^{1,2}(\mathbb{R}, M) = \{u \in W^{1,2}(\mathbb{R}, \mathbb{R}^N) : u(\mathbb{R}) \subset M\}.$$

To ensure that constant maps $u : \mathbb{R} \rightarrow p \in M$ belong to the model, define

$$W_{\text{loc}}^{1,2}(\mathbb{R}, M) = \{u : \mathbb{R} \rightarrow M : \forall K \subset \mathbb{R} \text{ compact, } u|_K \in W^{1,2}\}.$$

5.1 Functional Setting

Let M be a closed manifold of dimension m , let

$$f : M \rightarrow \mathbb{R}$$

be Morse, and let $p, q \in \text{Crit } f$. Define

$$\mathcal{P} = \left\{ u \in W_{\text{loc}}^{1,2}(\mathbb{R}, M) \left| \begin{array}{l} \exists s_0 > 0, \exists u_- \in W^{1,2}((-\infty, -s_0), T_p M), \\ \exists u_+ \in W^{1,2}((s_0, \infty), T_q M), \\ u(s) = \exp_p u_-(s) \text{ for } s < -s_0, \\ u(s) = \exp_q u_+(s) \text{ for } s > s_0 \end{array} \right. \right\}.$$

For $u \in \mathcal{P}$, define

$$\mathcal{E}_u = \{\text{the space of } L^2\text{-vector fields along } u\} = L^2(u^*TM).$$

Thus

$$\mathcal{E} \rightarrow \mathcal{P}$$

is a Banach bundle modeled on $W^{1,2}(\mathbb{R}, \mathbb{R}^n)$, with fiber $L^2(\mathbb{R}, \mathbb{R}^n)$.

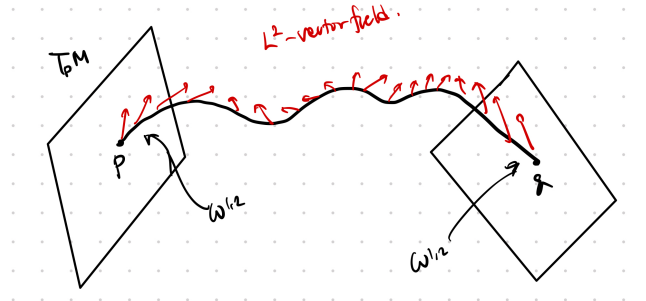


Figure 7: The Moduli space

Let g be a metric on M . Define

$$F_g(u) = \partial_s u + \nabla_g f(u)$$

in the L^2 sense. Then

$$F_g(u) = 0 \iff \partial_s u = -\nabla_g f(u).$$

The goal is

$$F_g^{-1}(0_{\mathcal{E}}) = \widehat{\mathcal{M}}(p, q),$$

and we want to give it a smooth manifold structure of dimension

$$\operatorname{ind}_f p - \operatorname{ind}_f q.$$

Let $\mathcal{M}$ denote the space of C^k metrics on M . Define

$$F : \mathcal{P} \times \mathcal{M} \rightarrow \mathcal{E}, \quad F(u, g) = \partial_s u + \nabla_g f(u).$$

Later, we will use the transversality theorem.

Lemma 1. *If $F(u, g) = 0$, then $u \in C^{k+1}$.*

Sketch. If $g \in C^k$ and $f \in C^\infty$, then $\nabla_g f \in C^k$. Since $u \in W^{1,2}$ and $u \in C^0$,

$$F(u, g) = 0 \Rightarrow \partial_s u = -\nabla_g f(u) \in C^0.$$

Thus $\partial_s u$ is the usual derivative of u , so $u \in C^1$. Inductively,

$$\partial_s u = -\nabla_g f(u) \in C^1$$

implies $u \in C^2$, and so on. Hence $u \in C^{k+1}$. □

Step 1. We apply the transversality theorem to show the manifold structure. Here $\mathcal{E} \rightarrow \mathcal{P} \times \mathcal{M}$ is a Banach vector bundle. Assume

$$(u, g) \in F^{-1}(0_{\mathcal{E}}).$$

The required conditions are

1. $T_{(u,g)}(\mathcal{P} \times \mathcal{M}) \rightarrow T_{((u,g),0)}\mathcal{E} \rightarrow \mathcal{E}_{(u,g)}$ is surjective,
2. $D_u F_g : T_u \mathcal{P} \rightarrow \mathcal{E}_{(u,g)}$ is Fredholm.

These are the two transversality-theorem hypotheses.

For $u \in \mathcal{P}$, choose a parallel transported basis

$$e_1, \dots, e_m$$

of u^*TM along u such that

$$\nabla_{\partial_s u} e_i = 0.$$

Then

$$T_u \mathcal{P} \cong W^{1,2}(\mathbb{R}, \mathbb{R}^m), \quad \mathcal{E}_u \cong L^2(\mathbb{R}, \mathbb{R}^m).$$

Writing

$$V = \sum_{i=1}^m V_i e_i \in T_u \mathcal{P},$$

and

$$\nabla_{e_j} \nabla_g f(u(s)) = \sum_{i=1}^m A_s^{ij} e_i(u(s)), \quad A_s = (A_s^{ij})_{1 \leq i, j \leq m},$$

we obtain the linearization.

The differential of F_g is

$$\begin{aligned} D_u F_g(V) &= \partial_s V + \nabla_V \nabla_g f(u) \\ &= \partial_s V + \sum_j V_j \nabla_{e_j} \nabla_g f(u) \\ &= \sum_i \left(\partial_s V_i + \sum_j V_j A_s^{ij} \right) e_i. \end{aligned}$$

Thus $D_u F_g$ has the form

$$W^{1,2}(\mathbb{R}, \mathbb{R}^m) \rightarrow L^2(\mathbb{R}, \mathbb{R}^m), \quad V \mapsto \partial_s V + A_s V.$$

Moreover,

$$\lim_{s \rightarrow -\infty} A(s) = \text{Hess}_g f(p), \quad \lim_{s \rightarrow +\infty} A(s) = \text{Hess}_g f(q).$$

Assume $s \mapsto A_s$ is C^k and that both limits are symmetric nonsingular matrices:

$$A_{\pm\infty} := \lim_{s \rightarrow \pm\infty} A_s.$$

Let

$$\lambda(A_{\pm\infty}) = \text{number of negative eigenvalues.}$$

Theorem 22. *The operator*

$$\partial_s + A_s : W^{1,2}(\mathbb{R}, \mathbb{R}^m) \rightarrow L^2(\mathbb{R}, \mathbb{R}^m)$$

is Fredholm of index

$$\lambda(A_{-\infty}) - \lambda(A_{+\infty}).$$

Lemma 2 (Closed range lemma). *Let A, B, C be Banach spaces, and let*

$$L : A \rightarrow B$$

be bounded and

$$K : A \rightarrow C$$

be bounded linear compact. Assume there is $d > 0$ such that for all $a \in A$,

$$\|a\|_A \leq d(\|La\|_B + \|Ka\|_C).$$

Then

$$\dim \ker L < \infty, \quad \text{im } L \text{ is closed.}$$

Sketch. If $\dim \ker L = \infty$, choose $a_n \in \ker L$ with $\|a_n\|_A = 1$. Since K is compact, Ka_n has a convergent subsequence. Then

$$\|a_n - a_m\|_A \leq d\|Ka_n - Ka_m\|_C \rightarrow 0,$$

so $\{a_n\}$ has a Cauchy subsequence, contradicting infinite-dimensionality.

For the closed image, since $\dim \ker L < \infty$, choose a closed complement A_0 with

$$A = \ker L \oplus A_0.$$

Suppose $La_n \rightarrow b$ with $a_n \in A_0$.

Claim. The sequence a_n is bounded. If not, put

$$\bar{a}_n = \frac{a_n}{\|a_n\|}.$$

Then

$$\|L\bar{a}_n\| = \frac{1}{\|a_n\|} \|La_n\| \rightarrow 0.$$

Since $K\bar{a}_n$ has a convergent subsequence and

$$\|\bar{a}_n - \bar{a}_m\|_A \leq d\|L\bar{a}_n - L\bar{a}_m\|_B + d\|K\bar{a}_n - K\bar{a}_m\|_C,$$

we obtain a subsequence converging to some $\bar{a} \in A_0$. Then $L\bar{a} = 0$, so $\bar{a} = 0$, a contradiction to

$$\|\bar{a}_n - \bar{a}\| = 1.$$

Thus a_n is bounded and Ka_n has a convergent subsequence. Again

$$\|a_n - a_m\|_A \leq d\|La_n - La_m\|_B + d\|Ka_n - Ka_m\|_C \rightarrow 0.$$

Hence a_n is Cauchy. If $a_n \rightarrow a$, then $b = La$, so $\text{im } L$ is closed. □

Now let $\mathcal{L}$ be the space of bounded linear operators

$$W^{1,2}(\mathbb{R}, \mathbb{R}^m) \rightarrow L^2(\mathbb{R}, \mathbb{R}^m).$$

Define

$$\mathcal{D} = \left\{ L_A = \partial_s + A_s \mid A_s : \mathbb{R} \rightarrow \mathbb{R}^{m \times m}, A_{\pm\infty} := \lim_{s \rightarrow \pm\infty} A_s \in GL(\mathbb{R}^m) \cap \text{Sym}(\mathbb{R}^m) \right\}.$$

Let

$$\mathcal{D}(L_A) = \{L_B : B_{\pm\infty} = A_{\pm\infty}\}.$$

Lemma 3. $\mathcal{D}(L_A)$ is contractible.

Indeed,

$$H : [0, 1] \times \mathcal{D}(L_A) \rightarrow \mathcal{D}(L_A), \quad (t, L_B) \mapsto L_{tA+(1-t)B},$$

is continuous in the operator norm.

Corollary 5. The Fredholm index is constant on $\mathcal{D}(L_A)$.

Proposition 3. Assuming L_A is Fredholm, one has

$$\text{ind } L_A = \lambda(A_-) - \lambda(A_+).$$

Step 1. Assume $A_\pm$ is diagonal. Choose $C_s \in SL(\mathbb{R}^m)$ such that

$$C_\pm A_\pm C_\pm^{-1} = D_\pm = \begin{pmatrix} \lambda_\pm^1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_\pm^m \end{pmatrix},$$

Set

$$C(s) = \begin{cases} C_-, & s < -s_0, \\ C_+, & s > s_0. \end{cases}$$

Then multiplication by C induces isomorphisms

$$C_{W^{1,2}} : W^{1,2}(\mathbb{R}, \mathbb{R}^m) \rightarrow W^{1,2}(\mathbb{R}, \mathbb{R}^m), \quad f \mapsto Cf,$$

and

$$C_{L^2} : L^2(\mathbb{R}, \mathbb{R}^m) \rightarrow L^2(\mathbb{R}, \mathbb{R}^m), \quad f \mapsto Cf.$$

Then

$$C_{L^2} L_A C_{W^{1,2}}^{-1} f = C_s (\partial_s + A_s) C_s^{-1} f = \partial_s f + (C_s \partial_s C_s^{-1} + C A_s C_s^{-1}) f = L_D f.$$

Because

$$\lim_{s \rightarrow \pm\infty} D(s) = D_\pm,$$

we get

$$L_D = C_{L^2} \circ L_A \circ C_{W^{1,2}}^{-1} \in \mathcal{D}(L_D)$$

and hence

$$\text{ind } L_D = \text{ind } L_A.$$

Step 2. Assume D_s is diagonal for every $s \in \mathbb{R}$. Choose

$$D_s = \begin{pmatrix} \lambda_s^1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_s^m \end{pmatrix}$$

where

$$\lambda_s^i = \begin{cases} \lambda_-^i, & s < -s_0, \\ \lambda_+^i, & s > s_0. \end{cases}$$

Then for

$$f = (f_1, \dots, f_m) \in \ker L_D \subset W^{1,2}(\mathbb{R}, \mathbb{R}^m),$$

we have

$$\partial_s f_i = -\lambda_s^i f_i.$$

Thus

$$f_i(s) = \begin{cases} e^{-\lambda_-^i(s+s_0)} f_i(-s_0), & s < -s_0, \\ e^{-\lambda_+^i(s-s_0)} f_i(s_0), & s > s_0. \end{cases}$$

Therefore f_i extends precisely when

$$\lambda_-^i < 0, \quad \lambda_+^i > 0.$$

Hence

$$\dim \ker L_D = |\{i : \lambda_-^i < 0, \lambda_+^i > 0\}|,$$

and

$$\dim \text{coker } L_D = \dim \ker(-\partial_s + D) = |\{i : \lambda_-^i > 0, \lambda_+^i < 0\}|.$$

Thus

$$\text{ind } L_D = \lambda(A_-) - \lambda(A_+).$$

□

Closed range lemma for $L = \partial_s + A_s$. Let

$$L = \partial_s + A_s : W^{1,2}(\mathbb{R}, \mathbb{R}^m) \rightarrow L^2(\mathbb{R}, \mathbb{R}^m),$$

and let

$$K : W^{1,2}(\mathbb{R}, \mathbb{R}^m) \rightarrow L^2([-s, s], \mathbb{R}^m)$$

be restriction.

Theorem 23. *For large $s > 0$, there is $c > 0$ such that for all $\psi \in W^{1,2}(\mathbb{R}, \mathbb{R}^m)$,*

$$\|\psi\|_{W^{1,2}} \leq c(\|L\psi\|_{L^2} + \|K\psi\|_{L^2}).$$

If this holds, then $\dim \ker L < \infty$ and $\text{im } L$ is closed by the closed range lemma.

Step 1: simple case. Assume $A = A_s \in \text{Sym} \cap GL$ is constant. Then $L = \partial_s + A$ is bijective. Write

$$\mathbb{R}^m = E_+ \oplus E_-,$$

where $A_\pm = A|_{E_\pm}$ and

$$P_\pm : \mathbb{R}^m \rightarrow E_\pm$$

are the projections. Define

$$K(s) = \begin{cases} e^{-A_+ s} P_+, & s > 0, \\ -e^{-A_- s} P_-, & s < 0. \end{cases}$$

Then

$$\|K\|_{L^1} < \infty$$

although K is discontinuous at $s = 0$. Define

$$Q : L^2(\mathbb{R}, \mathbb{R}^m) \rightarrow W^{1,2}(\mathbb{R}, \mathbb{R}^m), \quad \eta \mapsto K * \eta.$$

Claim. $L \circ Q = \text{id}$. Indeed, writing

$$Q\eta = \psi = \psi_+ + \psi_-,$$

we have

$$\psi_+ = \int_{-\infty}^s e^{-A(s-t)} \eta_+(t) dt, \quad \psi_- = - \int_s^\infty e^{-A(s-t)} \eta_-(t) dt.$$

Then

$$\partial_s \psi_+ = \eta_+ - A\psi_+, \quad \partial_s \psi_- = \eta_- - A\psi_-.$$

Thus

$$\partial_s \psi + A\psi = \eta_+ + \eta_- = \eta.$$

Claim. L is injective. By the $SL(m)$ argument we may assume A is diagonal. If

$$\xi \in \ker L,$$

then

$$0 = \partial_s \xi_i + \lambda_i \xi_i,$$

which gives $\xi_i \in L^2$ only when $\xi_i = 0$.

Claim. $K * \eta \in W^{1,2}(\mathbb{R}, \mathbb{R}^m)$. Since

$$\|K * \eta\|_{L^2}^2 = \int_{-\infty}^\infty \left| \int_{-\infty}^\infty K(s-t) \eta(t) dt \right|^2 ds$$

we get by Young's inequality

$$\|K * \eta\|_{L^2} \leq \|K\|_{L^1} \|\eta\|_{L^2}.$$

Moreover,

$$\partial_s(K * \eta) = \eta - A(K * \eta),$$

so

$$\|\partial_s(K * \eta)\|_{L^2} \leq \|\eta\|_{L^2} + \|A\| \|K * \eta\|_{L^2}.$$

Hence

$$\|K * \eta\|_{W^{1,2}} \leq C \|\eta\|_{L^2}.$$

□

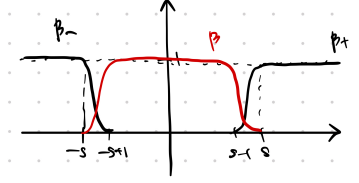


Figure 8: Cutoff function

Step 2: general case. Choose cutoff functions as in the sketches. For $s > S$, take A_s close to $A_{+\infty}$:

$$\|A_s - A_{+\infty}\| < \varepsilon.$$

Then

$$\|\beta - \xi\|_{W^{1,2}} \leq C\|(\partial_s + A_{+\infty})\beta - \xi\|_{L^2}$$

and hence

$$\begin{aligned} \|\beta\xi\|_{W^{1,2}} &\leq C(\|L(\beta\xi)\|_{L^2} + \|\partial_s\beta\xi\|_{L^2} + \|(A_s - A_{+\infty})\beta\xi\|_{L^2}) \\ &\leq C(\|L(\beta\xi)\|_{L^2} + \|\partial_s\beta\xi\|_{L^2} + \varepsilon\|\beta\xi\|_{L^2}). \end{aligned}$$

Therefore, for ε small,

$$\|\beta\xi\|_{W^{1,2}} \leq C(\|L(\beta\xi)\|_{L^2} + \|\partial_s\beta\xi\|_{L^2}).$$

The analogous estimate for the left end also holds. Combining with a middle cutoff β gives

$$\|(1 - \beta)\xi\|_{W^{1,2}} \leq C\|L((1 - \beta)\xi)\|_{L^2}$$

and

$$\|\beta\xi\|_{W^{1,2}} \leq C(\|\beta\xi\|_{L^2} + \|\partial_s(\beta\xi)\|_{L^2}).$$

The estimates imply

$$\|\xi\|_{W^{1,2}} \leq C(\|K\xi\|_{L^2} + \|L\xi\|_{L^2}).$$

□

Corollary 6. *The operator*

$$L = \partial_s + A_s$$

has finite-dimensional kernel and closed image.

The remaining part is to show

$$\dim \operatorname{coker} L < \infty.$$

The formal adjoint is

$$L^* = -\partial_s + A_s^* : W^{1,2}(\mathbb{R}, \mathbb{R}^m) \rightarrow L^2(\mathbb{R}, \mathbb{R}^m),$$

with

$$\langle x, Ly \rangle_{L^2} = \langle xL^*, y \rangle_{L^2}.$$

Lemma 4.

$$\operatorname{coker} L \cong \ker L^*.$$

Proof. Since $\operatorname{im} L$ is closed and $L^2(\mathbb{R}, \mathbb{R}^m)$ is Hilbert,

$$L^2(\mathbb{R}, \mathbb{R}^m) \cong \operatorname{im} L \oplus (\operatorname{im} L)^\perp.$$

Remember: closedness is needed for an orthogonal decomposition in a Hilbert space. Thus

$$\operatorname{coker} L = L^2(\mathbb{R}, \mathbb{R}^m) / \operatorname{im} L \cong (\operatorname{im} L)^\perp = \ker L^*.$$

If $y \in \ker L^{\operatorname{adj}}$, then $y \in W^{1,2}$ and

$$0 = \langle L^{\operatorname{adj}} u, y \rangle_{W^{1,2}} = \langle u, L^* y \rangle_{L^2}$$

for all u . Hence $y \in \ker L^*$, and conversely.

□

Therefore

$$\dim \operatorname{coker} L = \dim \ker L^* < \infty.$$

Recall the transversality theorem. Let $E \rightarrow M \times S$ be a Banach vector bundle with fiber F . Assume

$$(m, s) \in F^{-1}(0_E).$$

If

1. $T_{(m,s)}(M \times S) \rightarrow T_{(m,s,0)}E \rightarrow E_{(m,s)}$ is surjective,
2. $D_m F_s = \pi \circ d_{(m,s)} F : T_m M \rightarrow E_{(m,s)}$ is Fredholm,

then for generic $s \in S$, $D_m F_s$ is surjective and $F_s^{-1}(0_E)$ is a smooth manifold with

$$T_m(F_s^{-1}(0_E)) = \ker D_m F_s = \text{ind } D_m F_s.$$

Theorem 24. *There exists a Baire set of metrics*

$$\mathcal{M}_{\text{reg}} \subset \mathcal{M}$$

such that, for every $g \in \mathcal{M}_{\text{reg}}$,

$$F_g^{-1}(0_{\mathcal{E}}) = \widehat{\mathcal{M}}(p, q)$$

is a smooth manifold of dimension

$$\text{ind}_f p - \text{ind}_f q.$$

Proof. Recall

$$F(u, g) = \partial_s u + \nabla_g f(u),$$

and

$$\mathcal{E}_u = \{L^2 \text{ vector fields along } u\} = L^2(u^* TM).$$

The path space is

$$\mathcal{P} = \left\{ u \in W_{\text{loc}}^{1,2}(\mathbb{R}, M) \left| \begin{array}{l} \exists s_0 > 0, \exists u_- \in W^{1,2}((-\infty, -s_0), T_p M), \\ \exists u_+ \in W^{1,2}(s_0, \infty), T_q M), \\ u(s) = \exp_p u_-(s), \quad s < -s_0, \\ u(s) = \exp_q u_+(s), \quad s > s_0 \end{array} \right. \right\}.$$

Step 1. For every $(u, g) \in F^{-1}(0_{\mathcal{E}})$, the map

$$D_{(u,g)} F : T_u \mathcal{P} \times T_g \mathcal{M}^k \rightarrow \mathcal{E}_{(u,g)}$$

is surjective. We compute

$$D_{(u,g)} F(\hat{u}, \hat{g}) = D_u F_g(\hat{u}) + D_g \nabla_g f(\hat{g}).$$

Since $D_u F_g$ is Fredholm of index $\text{ind}_f p - \text{ind}_f q$, it remains to show surjectivity.

Claim 1.

$$\text{coker } D_{(u,g)} F \subset \ker(D_u F_g)^{\text{adj}} \cap (\text{im } D_g \nabla_g f)^{\perp}.$$

Indeed, since $D_u F_g$ is Fredholm, $\text{im } D_u F_g$ is closed in B and has finite codimension, so

$$B = \text{im } D_u F_g \oplus C$$

for a finite-dimensional complement C . Thus

$$\text{im } L = \text{im } D_u F_g \oplus (C \cap \text{im } L),$$

and $\text{im } L$ is closed. Since

$$\text{im } D_u F_g \subset \text{im } L,$$

we have

$$(\text{im } D_u F_g)^{\perp} \supset (\text{im } L)^{\perp} = \ker D_u F_g^{\text{adj}}.$$

Thus the cokernel lies in the indicated intersection.

Assume $D_{(u,g)} F$ is not surjective. Then there exists

$$0 \neq b \in \text{coker } D_{(u,g)} F \subset \mathcal{E}_u = L^2(u^* TM)$$

with

$$\langle D_g \nabla_g f(\hat{g}), b \rangle_{L^2} = 0$$

for all $\hat{g} \in T_g \mathcal{M}^k$. Since $b \neq 0$, $b \in W^{1,2} \subset C^0$, so for some s_0 ,

$$b(s_0) \neq 0.$$

Claim 2. There exists

$$\hat{g} \in T_{u(s_0)} \text{Sym}^2(T^*M)$$

such that

$$\langle D_g \nabla f|_{u(s_0)} \hat{g}, b(s_0) \rangle_{u(s_0)} > 0.$$

In a local coordinate chart,

$$\nabla_g f = g^{-1} \text{grad } f.$$

Therefore

$$D_g \nabla f(\hat{g})|_x = \frac{d}{dt} \Big|_{t=0} (g + t\hat{g})^{-1} (\text{grad } f)|_x = -g^{-1} \hat{g} g^{-1} (\text{grad } f)|_x.$$

Choose a symmetric matrix S such that

$$S(\text{grad } f|_x) = b(s_0).$$

Let

$$\tilde{g} = -gSg.$$

Then

$$\langle D_g \nabla f|_{u(s_0)} \tilde{g}, b(s_0) \rangle_{u(s_0)} = S(\text{grad } f|_x, b(s_0)) = |b(s_0)|^2 > 0.$$

Take any $\hat{g} \in T_g \mathcal{M}^k$ such that

$$\hat{g}|_{u(s_0)} = \tilde{g}.$$

Then

$$\langle D_g \nabla f(\hat{g}), b \rangle > 0$$

for some neighborhood of $u(s_0)$. Choosing a bump function

$$\beta : M \rightarrow [0, 1], \quad \text{supp } \beta \subset U,$$

and replacing $\hat{g}$ by $\beta \hat{g}$ gives

$$\langle D_g \nabla f(\beta \hat{g}), b \rangle_{L^2} > 0,$$

which is a contradiction. Hence $D_{(u,g)} F$ is surjective.

Step 2. By the transversality theorem, there exists a Baire set

$$\mathcal{M}_{\text{reg}}^k \subset \mathcal{M}^k$$

such that for all $g \in \mathcal{M}_{\text{reg}}^k$, $D_u F_g$ is surjective. Since C^∞ is dense and transversality is an open condition,

$$\mathcal{M}_{\text{reg}} = \bigcap_k \mathcal{M}_{\text{reg}}^k$$

is a Baire set. Therefore, for $g \in \mathcal{M}_{\text{reg}}$,

$$F_g^{-1}(0_\mathcal{E})$$

is a smooth manifold of index

$$\text{ind } D_u F_g = \text{ind}_f p - \text{ind}_f q.$$

□

5.2 The Identity $\partial \circ \partial = 0$

Let $f : M \rightarrow \mathbb{R}$ be Morse, $p, q \in \text{Crit } f$, and let g be a generic Riemannian metric. We want to show that

$$\widehat{\mathcal{M}}(p, q)$$

has manifold structure and that $\partial \circ \partial = 0$.

For a path $u : \mathbb{R} \rightarrow M$, define the energy

$$E(u) = \int_{-\infty}^{\infty} \|\partial_s u\|^2 ds.$$

If $\partial_s u = -\nabla_g f(u)$, then

$$\begin{aligned} E(u) &= \int_{-\infty}^{\infty} g(\partial_s u, \partial_s u) ds = \int_{-\infty}^{\infty} \langle \partial_s u, -\nabla_g f(u) \rangle_g ds \\ &= - \int_{-\infty}^{\infty} \frac{d}{ds} f(u(s)) ds = f(u(-\infty)) - f(u(+\infty)). \end{aligned}$$

Since $f \circ u$ is monotone, this limit exists.

Consequently, if

$$f(p) < f(q),$$

then

$$\widehat{\mathcal{M}}(p, q) = \emptyset,$$

while

$$\widehat{\mathcal{M}}(p, p) = \{u_p\}.$$

Lemma 5. *If $\partial_s u = -\nabla_g f(u)$ and $\lim_{s \rightarrow -\infty} u(s) \in \text{Crit } f$, then the limit is a critical point.*

Sketch. Let $p \in A_p \subset B_p$ with

$$\overline{B_p} \cap \text{Crit } f = \{p\}.$$

For a compact interval $I = [a, b]$, if $u(\partial I) \in \partial B_p$ and $u(I)$ meets A_p , then

$$E(u|_I) > d$$

for a positive constant d . Indeed, set

$$k = \min_{B_p \setminus A_p} \|\nabla_g f\|_g > 0.$$

Then

$$E(u|_{[a,b]}) = \|\partial_s u\|_{L^2([a,b])}^2 \geq k \|\partial_s u\|_{L^1(I)}^2 \geq k \text{dist}(A_p, B_p^c)^2.$$

Since $\text{Crit } f$ is finite, choose d satisfying the claim for all critical points. □

Since

$$E(u) \leq \max f - \min f,$$

there exists $s_0 > 0$ such that

$$E(u|_{[s_0, \infty)}) < d.$$

If $s_1 > s_0$ and $u(s_1) \notin \bigcup_{p \in \text{Crit } f} B_p$, then, if $u(s_2) \in \bigcup \partial A_p$ for some critical point, a contradiction follows. Hence

$$u([s_1, \infty)) \notin \bigcup B_p$$

would imply

$$\inf_{M \setminus \bigcup B_p} \|\nabla_g f\|_g > d,$$

another contradiction. Therefore for $s \geq s_0$, $u(s) \in B_p$ for some $p \in \text{Crit } f$. Shrink B_p to get convergence.

Define

$$\widehat{\mathcal{M}}(p, q) \in \mathcal{P} \subset W_{\text{loc}}^{1,2}(\mathbb{R}, M) \subset C_{\text{loc}}^0(\mathbb{R}, M).$$

Set

$$\mathcal{M}(p, q) = \widehat{\mathcal{M}}(p, q)/\mathbb{R}.$$

Give $\mathcal{P}$ a topology by declaring

$$[u_n] \rightarrow [u]$$

iff there exist $s_n \in \mathbb{R}$ such that

$$u_n(\cdot + s_n) \xrightarrow{C_{\text{loc}}^\infty} u.$$

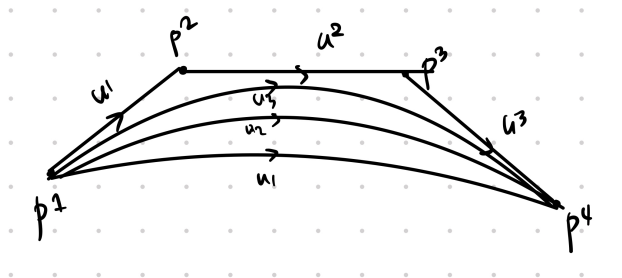


Figure 9: The Convergence

Lemma 6. *The union*

$$\widehat{\mathcal{M}} = \bigcup_{p,q \in \text{Crit } f} \widehat{\mathcal{M}}(p,q)$$

is C_{loc}^∞ -compact.

Sketch. Let $(u_n) \subset \widehat{\mathcal{M}}$. For every compact interval $[-s, s] \subset \mathbb{R}$, $u_n|_{[-s,s]}$ is uniformly bounded and equicontinuous because

$$\text{dist}_g(u_n(s_0), u_n(s_1)) \leq \int_{s_0}^{s_1} \|\partial_s u_n\| ds \leq |s_1 - s_0| \max_{x \in M} \|\nabla_g f(x)\|.$$

By Arzela-Ascoli,

$$u_n \rightarrow u \quad \text{in } C_{\text{loc}}^0.$$

Then

$$\nabla_g f(u_n) \rightarrow \nabla_g f(u)$$

in C_{loc}^0 , and using

$$\partial_s u_n = -\nabla_g f(u_n)$$

gives $u \in C^1$. Iterating gives $u \in C^\infty$, and

$$u_n \rightarrow u \quad \text{in } C_{\text{loc}}^\infty.$$

If $u_n \rightarrow u$ in C_{loc}^∞ in $\widehat{\mathcal{M}}(p, q)$, then it also converges in $W^{1,2}$. □

5.3 Compactness and Gluing in Morse Homology

Theorem 25 (Compactness). *Let*

$$(u_k) \subset \widehat{\mathcal{M}}(p_-, p_+).$$

Then there exist critical points

$$p^1 = p_-, \dots, p^{\ell+1} = p_+ \in \text{Crit } f,$$

and trajectories

$$u^1 \in \widehat{\mathcal{M}}(p^1, p^2), \dots, u^\ell \in \widehat{\mathcal{M}}(p^\ell, p^{\ell+1}),$$

and sequences

$$(s_k^1), \dots, (s_k^\ell) \subset \mathbb{R}$$

such that

$$(s_k^j)_* u_k \xrightarrow{C_{\text{loc}}^\infty} u^j \quad \text{as } k \rightarrow \infty.$$

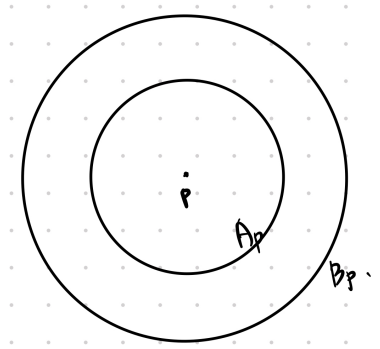


Figure 10: A_p and B_p

Proof. Step 1. Set $j = 1$. Choose an open neighborhood V of p^1 such that

$$\overline{V} \cap \text{Crit } f = \{p^1\}.$$

Define

$$s_k^1 = \inf\{s \in \mathbb{R} \mid u_k(s) \notin V\} < \infty.$$

Passing to a subsequence,

$$(s_k^1)_* u_k \xrightarrow{C_{\text{loc}}^\infty} u^1 \in \widehat{\mathcal{M}}$$

by the compactness lemma. Since $\widehat{\mathcal{M}}$ is C_{loc}^∞ compact, this limit exists after passing to a subsequence. Moreover,

$$u^1(0) = \lim_{k \rightarrow \infty} u_k(s_k^1) \in \partial V,$$

so u^1 is not constant. If $s < 0$, then

$$(s_k^1)_* u_k(s) \in \overline{V},$$

and hence

$$u^1(s) \in \overline{V}, \quad \lim_{s \rightarrow -\infty} u^1(s) = p^1 = p_-.$$

If

$$\lim_{s \rightarrow \infty} u^1(s) = p_+,$$

then $\ell = 1$ and the proof is complete.

Step 2. Set $j = 2$. Let

$$p^2 = \lim_{s \rightarrow \infty} u^1(s).$$

Let B_δ be the δ -ball centered at p^2 and choose $\delta > 0$ so that

$$\overline{B_\delta} \cap \text{Crit } f = \{p^2\}.$$

Then there exists $s_0 \in \mathbb{R}$ such that

$$u^1([s_0, \infty)) \subset B_{\delta/2}.$$

Since

$$(s_k^1)_* u_k \xrightarrow{C_{\text{loc}}^\infty} u^1,$$

we have, for k sufficiently large,

$$(s_k^1)_* u_k(s_0) \in B_\delta.$$

Define

$$R_k = \inf\{r > 0 \mid (s_k^1)_* u_k(s_0 + r) \notin B_\delta\} < \infty.$$

Here finiteness follows since

$$\lim_{s \rightarrow \infty} u_k(s + s_0 + s_k^1) = p_+$$

for every k . We have

$$\lim_{k \rightarrow \infty} R_k = \infty.$$

Indeed, if R_k were bounded, then $(s_k^1)_* u_k$ would not converge to u^1 in C_{loc}^∞ .

Define

$$s_k^2 = s_k^1 + R_k + s_0.$$

After passing to a subsequence,

$$(s_k^2)_* u_k \longrightarrow u^2 \in \widehat{\mathcal{M}}.$$

Then

$$u^2(0) \in \partial \overline{B_\delta},$$

so u^2 is not constant.

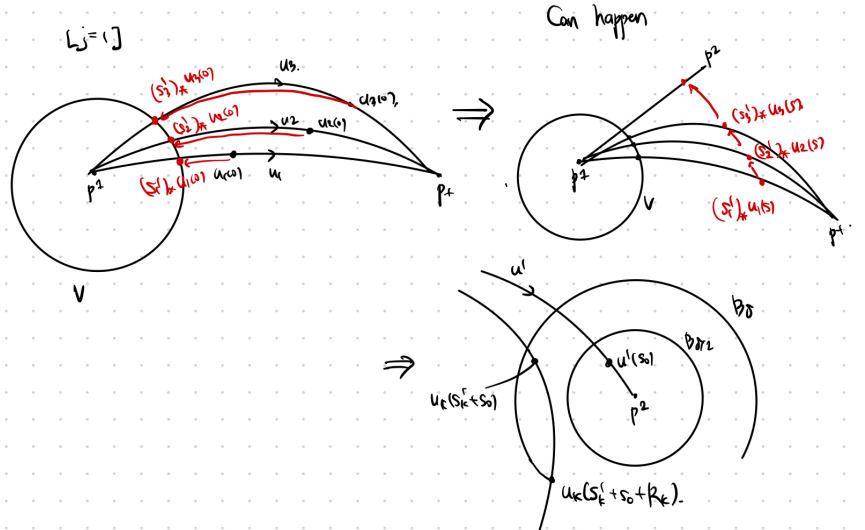


Figure 11: Step 1 and Step 2

We claim that

$$\lim_{s \rightarrow -\infty} u^2(s) = p^2.$$

Suppose instead that

$$q := \lim_{s \rightarrow -\infty} u^2(s) \neq p^2.$$

Let V_δ be the δ -ball centered at q such that

$$V_\delta \cap B_\delta = \emptyset.$$

Then there exists $s_1 < 0$ such that

$$u^2((-\infty, s_1]) \subset V_{\delta/2}.$$

Since

$$(s_k^2)_* u_k \xrightarrow{C_{\text{loc}}^\infty} u^2,$$

there exists $k_0 \in \mathbb{N}$ such that for all $k \geq k_0$,

$$(s_k^2)_* u_k(s_1) \in V_\delta.$$

On the other hand,

$$(s_k^1)_* u_k((s_0, s_0 + R_k)) \subset B_\delta,$$

so for every $s \in (-R_k, 0)$,

$$(s_k^2)_* u_k(s) \in B_\delta.$$

Since $R_k \rightarrow \infty$, we can take k sufficiently large so that

$$-R_k < s_1 < 0.$$

Then

$$(s_k^2)_* u_k(s_1) \in V_\delta \cap B_\delta = \emptyset,$$

a contradiction. Therefore

$$\lim_{s \rightarrow -\infty} u^2(s) = p^2.$$

Repeating the argument, set

$$p^{n+1} = \lim_{s \rightarrow \infty} u^n(s).$$

Since

$$f(p^1) > f(p^2) > \dots$$

and $\text{Crit } f$ is finite, this process terminates. At the final trajectory u^ℓ , one must have

$$\lim_{s \rightarrow \infty} u^\ell(s) = p_+.$$

If the process terminated at a point other than p_+ , then the last trajectory would have to be constant, contradicting the construction of the nonconstant limits above. Hence the desired broken trajectory is obtained. $\square$

This compactness result gives a topology on

$$\mathcal{M} = \bigcup_{p, q} \mathcal{M}(p, q)$$

more precisely, on the union of unbroken and broken trajectories

$$\mathcal{M} \cup \mathcal{M}^2 \cup \dots \cup \mathcal{M}^K.$$

We say that

$$[u_n] \in \mathcal{M}(p_-, p_+)$$

converges to

$$([u^1], \dots, [u^\ell]) \in \mathcal{M}(p_-, p^2) \times \dots \times \mathcal{M}(p^\ell, p_+)$$

if the sequences of shifts s_k^j as above exist.

We want to show that, in this topology, when

$$\text{ind } p - \text{ind } q = 2,$$

one has

$$\overline{\mathcal{M}(p, q)} = \mathcal{M}(p, q) \cup \bigcup_{\text{ind } r = \text{ind } p - 1} \mathcal{M}(p, r) \times \mathcal{M}(r, q).$$

The compactness theorem gives

$$\partial \overline{\mathcal{M}(p, q)} \subset \bigcup_r \mathcal{M}(p, r) \times \mathcal{M}(r, q).$$

Theorem 26 (Gluing). *Let $p, r, q \in \text{Crit } f$ satisfy*

$$\text{ind}_f p = \text{ind}_f r + 1 = \text{ind}_f q + 2.$$

Then there exists a gluing map

$$\# : \mathcal{M}(p, r) \times \mathcal{M}(r, q) \times (\lambda_0, \infty) \longrightarrow \mathcal{M}(p, q)$$

given by

$$([u], [w], \lambda) \longmapsto [u \#_\lambda w],$$

which satisfies the following properties:

1. $\#$ is a smooth embedding.

2.

$$[u \#_\lambda w] \longrightarrow ([u], [w]) \quad \text{as } \lambda \rightarrow \infty,$$

with respect to the topology defined above.

3. If $[v_n] \in \mathcal{M}(p, q)$ and

$$[v_n] \longrightarrow ([u], [w]),$$

then for all sufficiently large n there exists $\lambda_n \rightarrow \infty$ such that

$$[v_n] = [u \#_{\lambda_n} w].$$

In particular, the third condition implies that only one end point occurs near the broken trajectory.

Corollary 7. *For $\text{ind } p - \text{ind } q = 2$,*

$$\partial \overline{\mathcal{M}(p, q)} \supset \bigcup_r \mathcal{M}(p, r) \times \mathcal{M}(r, q).$$

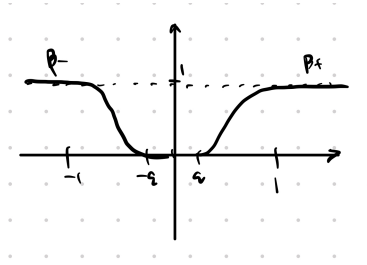


Figure 12: The Cutoff function

Sketch of gluing. For large λ , define a pre-glued path Z_λ by

$$Z_\lambda(s) = \begin{cases} u(s + \lambda), & s \leq -1, \\ \exp_r(\beta(-s) \exp_r^{-1}(u(s + \lambda)) + \beta(s) \exp_r^{-1}(w(s - \lambda))), & -1 \leq s \leq 1, \\ w(s - \lambda), & s \geq 1. \end{cases}$$

The problem is that

$$(\partial_s + A_s)Z_\lambda(s) \neq 0.$$

Nevertheless,

1. $Z_\lambda(s) = r$ for $s \in (-\epsilon, \epsilon)$.

2. $Z_\lambda(s) = u(s + \lambda)$ for $s \leq -1$, and hence

$$(-\lambda)_* Z_\lambda \xrightarrow{C_{\text{loc}}^\infty} u \quad \text{as } \lambda \rightarrow \infty.$$

3.

$$Z_\lambda \longrightarrow r \quad \text{as } \lambda \rightarrow \infty,$$

and

$$\lambda_* Z_\lambda \xrightarrow{C_{\text{loc}}^\infty} w \quad \text{as } \lambda \rightarrow \infty.$$

Let

$$b : Z_\lambda^* TM \longrightarrow (\exp_{Z_\lambda}(b))^* TM$$

be parallel transport. Consider the local expression

$$P^{-1} \circ \Phi_g \circ \exp_{Z_\lambda} : W^{1,p}(Z_\lambda^* TM) \longrightarrow L^p(Z_\lambda^* TM).$$

After trivialization, write

$$F_\lambda : W^{1,p}(\mathbb{R}, \mathbb{R}^m) \longrightarrow L^p(\mathbb{R}, \mathbb{R}^m).$$

Then

$$D_0 F_\lambda = \partial_s + A_\lambda(s),$$

where

$$A_\lambda(s) = \begin{cases} A_u(s + \lambda), & s \leq -1, \\ A_w(s - \lambda), & s \geq 1. \end{cases}$$

Thus $D_0 F_\lambda$ is Fredholm with index

$$\text{ind}_f p - \text{ind}_f q = 2.$$

Moreover,

$$\ker D_u F_\lambda \oplus \ker D_w F_\lambda \cong \ker D_0 F_\lambda.$$

Hence $D_0 F_\lambda$ is surjective, and a right inverse exists. **This is hard part.** □

Lemma 7. *Let $f : E \rightarrow F$ be a C^1 map between Banach spaces. Assume that df has a bounded right inverse G . Suppose*

$$\|f(0)\| < \frac{\epsilon}{4\|G\|},$$

where $\epsilon > 0$ is sufficiently small, and

$$\|d_\xi f - d_0 f\| < \frac{1}{2\|G\|} \quad \text{for every } \xi \in B_\epsilon(0).$$

Then there exists a unique

$$\xi \in B_\epsilon(0) \cap \text{im } G$$

such that

$$f(\xi) = 0, \quad \|\xi\| < 2\|Gf(0)\|.$$

Applying the lemma to $f = F_\lambda$, there exists $\lambda_0 > 0$ such that for every $\lambda \geq \lambda_0$ there exists a unique

$$\xi_\lambda \in \text{im } Q \cap B_\epsilon(0)$$

with

$$F_\lambda(\xi_\lambda) = 0, \quad \xi_\lambda \xrightarrow[\lambda \rightarrow \infty]{W^{1,p}} 0.$$

Define

$$u \#_\lambda w = \exp_{Z_\lambda}(\xi_\lambda).$$

Theorem 27. *In the Morse chain complex $MC(f, g)$,*

$$\partial \circ \partial = 0.$$

Proof. For a generator p ,

$$\partial p = \sum_r \# \mathcal{M}(p, r) r.$$

Therefore

$$\begin{aligned} \partial \circ \partial(p) &= \partial \left(\sum_r \# \mathcal{M}(p, r) r \right) \\ &= \sum_{r, q} \# \mathcal{M}(p, r) \# \mathcal{M}(r, q) q. \end{aligned}$$

By compactness and gluing,

$$\sum_r \# \mathcal{M}(p, r) \# \mathcal{M}(r, q) = \# \left(\overline{\partial \mathcal{M}(p, q)} \right) = 0.$$

Thus $\partial^2 = 0$. □

This shows that Morse homology in the modern viewpoint indeed defines a chain complex.

Theorem 28. *Let (f_0, g_0) and (f_1, g_1) be Morse–Smale pairs on M . Then there exists an isomorphism*

$$(\varphi_{10})_* : MH_*(f_0, g_0) \longrightarrow MH_*(f_1, g_1)$$

which satisfies

$$(\varphi_{21})_* \circ (\varphi_{10})_* = (\varphi_{20})_*, \quad (\varphi_{00})_* = \text{id}.$$

Proof. Let

$$f_s = \begin{cases} f_0, & s \leq -1, \\ f_1, & s \geq 1, \end{cases} \quad g_s = \begin{cases} g_0, & s \leq -1, \\ g_1, & s \geq 1. \end{cases}$$

Let $p_0 \in \text{Crit } f_0$ and $p_1 \in \text{Crit } f_1$. Define

$$C(p_0, p_1) = \left\{ u : \mathbb{R} \rightarrow M \mid \partial_s u + \nabla_{g_s} f_s(u) = 0, \lim_{s \rightarrow -\infty} u(s) = p_0, \lim_{s \rightarrow \infty} u(s) = p_1 \right\}.$$

Time-varying vector field.

The space $C(p_0, p_1)$ is a smooth manifold of dimension

$$\text{ind}_{f_0}(p_0) - \text{ind}_{f_1}(p_1).$$

Again, the transversality theorem and local coordinates using parallel transport give an operator of the form

$$\partial_s + A_s.$$

Its Fredholm index depends only on the boundary points.

For example, consider

$$\mathcal{P} = \{ u : \mathbb{R} \rightarrow M \mid u((-\infty, -s_0)) - p_0 \in W^{1,2}, u((s_0, \infty)) - p_1 \in W^{1,2} \}$$

and

$$\mathcal{M}^{1,2} = \{ g_s \mid g_s : \mathbb{R} \rightarrow C^\ell(T^*M), g_s = g_0 \text{ for } s \leq -1, g_s = g_1 \text{ for } s \geq 1 \}.$$

Define the continuation map

$$\varphi_{10} : MC_*(f_0, g_0) \longrightarrow MC_*(f_1, g_1)$$

by

$$p_0 \longmapsto \sum_{p_1} \#C(p_0, p_1) p_1,$$

where the sum is over $p_1 \in \text{Crit } f_1$ satisfying

$$\text{ind}_{f_0} p_0 = \text{ind}_{f_1} p_1.$$

Claim 1. φ_{10} is a chain map, i.e.

$$\partial_{f_1} \circ \varphi_{10} = \varphi_{10} \circ \partial_{f_0}.$$

Indeed,

$$\partial \overline{C(p_0, q_1)} = \left(\bigcup_{q_0} \mathcal{M}(p_0, q_0) \times C(q_0, q_1) \right) \cup \left(\bigcup_{p_1} C(p_0, p_1) \times \mathcal{M}(p_1, q_1) \right).$$

Here $\text{ind}_{f_1} q_1 = \text{ind}_{f_0} p_0 - 1$.

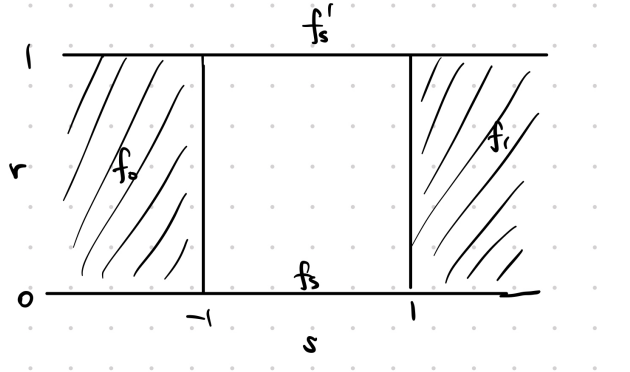


Figure 13: $f_{r,s}$

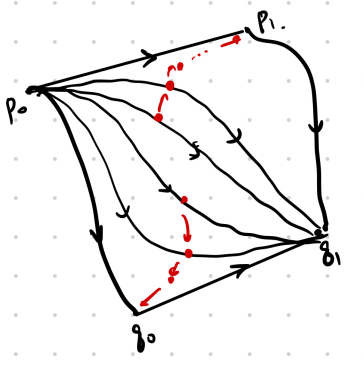


Figure 14: The Breaking

Thus the signed boundary count gives the chain map identity.

Claim 2. The induced map $(\varphi_{10})_*$ is independent of the choice of the homotopy (f_s, g_s) .

Let (f'_s, g'_s) be another homotopy, and let φ'_{10} be the corresponding continuation map. We want to construct a chain homotopy H such that

$$\varphi_{10} - \varphi'_{10} = H \circ \partial_0 + \partial_1 \circ H.$$

$$\begin{array}{ccccc} MC_{*+1}(f_0, g_0) & \xrightarrow{\partial_0} & MC_*(f_0, g_0) & \xrightarrow{\partial_0} & MC_{*-1}(f_0, g_0) \\ \downarrow \varphi_{10} & \swarrow H & \downarrow \varphi_{10} & \swarrow H & \downarrow \varphi_{10} \\ MC_{*+1}(f_1, g_1) & \xrightarrow{\partial_1} & MC_*(f_1, g_1) & \xrightarrow{\partial_1} & MC_{*-1}(f_1, g_1) \end{array}$$

Choose a homotopy

$$(f_s^r, g_s^r)_{r \in [0,1]}$$

with

$$(f_s^0, g_s^0) = (f_s, g_s), \quad (f_s^1, g_s^1) = (f'_s, g'_s).$$

For $p_0 \in \text{Crit } f_0$ and $p_1 \in \text{Crit } f_1$, define

$$\mathcal{H}(p_0, p_1) = \left\{ (r, u) \mid 0 \leq r \leq 1, \partial_s u + \nabla_{g_s^r} f_s^r(u) = 0, \lim_{s \rightarrow -\infty} u(s) = p_0, \lim_{s \rightarrow \infty} u(s) = p_1 \right\}.$$

Then $\mathcal{H}(p_0, p_1)$ is a smooth manifold of dimension

$$\text{ind}_{f_0} p_0 + 1 - \text{ind}_{f_1} p_1.$$

For $\text{ind}_{f_0} p_0 = \text{ind}_{f_1} p_1$, its boundary is

$$\begin{aligned} \partial \mathcal{H}(p_0, p_1) &= \mathcal{H}(p_0, p_1)|_{r=0} \cup \mathcal{H}(p_0, p_1)|_{r=1} \\ &\cup \bigcup_{q_0} \mathcal{M}(p_0, q_0) \times \mathcal{H}(q_0, p_1) \\ &\cup \bigcup_{q_1} \mathcal{H}(p_0, q_1) \times \mathcal{M}(q_1, p_1). \end{aligned}$$

The components at $r = 0$ and $r = 1$ correspond to φ_{10} and φ'_{10} , respectively, and the remaining components give

$$H \circ \partial_0 + \partial_1 \circ H.$$

Thus

$$\varphi_{10} - \varphi'_{10} = H \circ \partial_0 + \partial_1 \circ H,$$

so the induced map on homology is independent of the chosen homotopy.

Finally,

$$(\varphi_{21})_* \circ (\varphi_{10})_* = (\varphi_{20})_*.$$

Indeed, concatenate the homotopy from f_0 to f_1 with the homotopy from f_1 to f_2 . Gluing gives a homotopy from f_0 to f_2 , inducing a continuation map φ'_{20} . Since the induced continuation map is independent of the choice of homotopy,

$$(\varphi_{20})_* = (\varphi'_{20})_* = (\varphi_{21})_* \circ (\varphi_{10})_*.$$

Also, the constant homotopy gives

$$(\varphi_{00})_* = \text{id}.$$

Therefore Morse homology is independent of the Morse–Smale pair. $\square$

6 Local Coefficients

Let X be path connected, and let

$$\tilde{X} \rightarrow X$$

be the universal cover with fundamental group π . Then

$$\pi \curvearrowright \tilde{X}, \quad \pi \curvearrowright C_n(\tilde{X}),$$

and $C_n(\tilde{X})$ is a module over $\mathbb{Z}[\pi]$. The boundary

$$\partial : C_n(\tilde{X}) \rightarrow C_{n-1}(\tilde{X})$$

is a $\mathbb{Z}[\pi]$ -module homomorphism.

Let M be a module over $\mathbb{Z}[\pi]$. Define chains with local coefficients by

$$C_n(X; M) = C_n(\tilde{X}) \otimes_{\mathbb{Z}[\pi]} M,$$

with boundary

$$\partial \otimes_{\pi} 1.$$

Define cochains by

$$C^n(X; M) = \text{Hom}_{\mathbb{Z}[\pi]}(C_n(\tilde{X}), M).$$

Example. If M is the trivial $\mathbb{Z}[\pi]$ -module, then

$$\gamma m = m, \quad \gamma \in \pi,$$

and if $\sigma : \Delta^n \rightarrow X$ lifts to $\tilde{\sigma} : \Delta^n \rightarrow \tilde{X}$, then

$$\tilde{\sigma} \otimes m = \gamma \tilde{\sigma} \otimes m$$

so

$$C_n(\tilde{X}) \otimes_{\pi} M = C_n(X) \otimes M.$$

This is ordinary homology with coefficients in M .

Example. If $M = \mathbb{Z}[\pi]$ by its ring structure, then

$$C_n(\tilde{X}) \otimes_{\pi} \mathbb{Z}[\pi] \cong C_n(\tilde{X}), \quad H_n(X; \mathbb{Z}[\pi]) \cong H_n(\tilde{X}).$$

Geometrically, a local system can be interpreted as a bundle of groups

$$p : E \rightarrow X$$

with group structure on each fiber $p^{-1}(x)$, and local trivializations preserving the group structure. For homology with coefficients in a bundle E , a singular chain is a sum

$$\sum_i m_i \sigma_i,$$

where $\sigma_i : \Delta^n \rightarrow X$ and $m_i : \Delta^n \rightarrow E$ is a lifting. If n_i is another lifting of σ_i , then

$$(m_i + n_i)(s) = m_i(s) + n_i(s)$$

using the group operation in the fibers. The boundary is

$$\partial \left(\sum_i m_i \sigma_i \right) = \sum_j (-1)^j m_i \sigma_i|_{[v_0, \dots, \hat{v}_j, \dots, v_n]}.$$

7 Morse Homology with $\mathbb{Z}$ -Coefficients

Let V be a finite-dimensional vector space over $\mathbb{R}$. Its orientation line is

$$|V| = \mathbb{Z}\langle o_V, -o_V \rangle / (o_V + (-o_V) = 0).$$

The orientation of unstable manifolds of critical points determines

$$O_q = |T_q W^u(q)|.$$

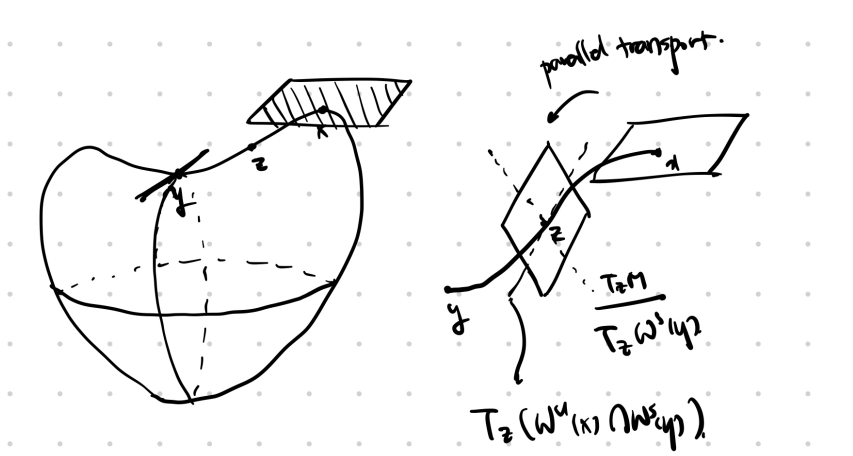


Figure 15: Morse Theory with orientation

Parallel transport and intersections give canonical identifications

$$\begin{aligned} T_x W^u(x) &\cong T_z W^u(x) \\ &\cong T_z(W^u(x) \cap W^s(y)) \oplus T_z M / T_z W^s(y) \\ &\cong T_z \mathcal{M}(x, y) \oplus \mathbb{R}\langle \dot{\gamma} \rangle \oplus O_y. \end{aligned}$$

Thus orienting the trajectory by $\partial_s \gamma$ specifies O_x and O_y , and induces an orientation on $T_\gamma \mathcal{M}(x, y)$.
If

$$\text{ind}(x) - \text{ind}(y) = 1,$$

then

$$O_x \cong \mathbb{R}\langle \dot{\gamma} \rangle \otimes O_y$$

and hence the trajectory induces an isomorphism

$$\tau_\gamma : O_x \rightarrow O_y.$$

Define

$$MC_i^{\mathbb{Z}}(f) = \bigoplus_{x \in \text{Crit } f} O_x,$$

with

$$\partial : MC_i^{\mathbb{Z}}(f) \rightarrow MC_{i-1}^{\mathbb{Z}}(f),$$

where

$$\partial|_{O_x} = \sum_{\substack{y \in \text{Crit } f \\ \text{ind } y = i-1}} \sum_{[\gamma] \in \mathcal{M}(x, y)} \tau_\gamma.$$

Why $\partial \circ \partial = 0$. Gluing gives

$$T_u \widehat{\mathcal{M}}(x, z) \oplus T_v \widehat{\mathcal{M}}(z, y) \longrightarrow T_w \widehat{\mathcal{M}}(x, y).$$

The vector fields $\partial_s u$ and $\partial_s v$ map to $\partial_s w$; $-\partial_s u$ and $\partial_s v$ correspond to the inward normal vector. If $w(0)$ is close enough to $u(0)$ and $w(R)$ close enough to $v(0)$, then the glued curve spends less time near z and the gluing parameter controls the end.

7.1 Local Coefficients Revisited

A local coefficient system L on X consists of

- a collection of R -modules $\{M_x\}_{x \in X}$,
- a collection of isomorphisms

$$T_{[\alpha]} : M_x \rightarrow M_y$$

for homotopy classes of paths $[\alpha]$, satisfying

$$T_{[\alpha]} = \text{id}_{M_x}, \quad T_{[\alpha][\beta]} = T_{[\alpha]} \circ T_{[\beta]}.$$

Another interpretation is that

$$\{T_{[\alpha]} : [\alpha] \in \pi_1(X, x)\}$$

gives automorphisms $M_x \rightarrow M_x$. Thus

$$\rho_x \in \text{Hom}(\pi_1(X, x), \text{Aut}(M_x)).$$

Fixing an isomorphism $M_x \cong M$, we have

$$\rho_x \in \text{Hom}(\pi_1(X, x), \text{Aut}(M)) / \text{Aut}(M),$$

because a change of identification acts by conjugation. Hence M is a right $\mathbb{Z}[\pi_1(X, x)]$ -module with

$$m \cdot g = \rho(g^{-1})(m).$$

Conversely, for a representation ρ , let

$$\tilde{X} \times_{\pi_1(X, x)} M \rightarrow X$$

be the associated bundle. An element of $\pi_1(X, x)$ acts as a deck transformation on $\tilde{X}$ and acts on M by ρ . Then any path from x to y lifts to an isomorphism

$$T_{[\alpha]} : M_x \rightarrow M_y.$$

Thus $C_*(\tilde{X})$ is a singular chain group with $\mathbb{Z}[\pi_1(X)]$ -action, and

$$H_*(C_*(\tilde{X}) \otimes_{\mathbb{Z}[\pi_1(X)]} M, \partial \otimes \text{id}) = H_*(X; L).$$

8 Poincare Duality

There is an isomorphism of Morse complexes

$$MC_*(-f) \cong MC^{m-*}(f),$$

so

$$MH_*(-f) \cong MH^{m-*}(f).$$

With local coefficients, let O^M be an orientation local system. Define

$$O_x^u(f) := |T_x W^u(x; -\nabla f)|, \quad O_x^s(f) := |T_x W^s(x; -\nabla f)|.$$

There is a canonical identification

$$O_x^u(f) \otimes O_x^s(f) \cong O_x^M.$$

Define

$$MC^*(f; O^M) = \text{Hom}_{\mathbb{Z}}(MC_*(f), O^M) = \bigoplus_{x \in \text{Crit } f} O_x^M \otimes (O_x^u(f))^{\vee}.$$

The cochain differential is induced by the maps

$$\tau_{\gamma} : O_x^u(-f) \rightarrow O_y^u(-f), \quad \tau_{\gamma} : O_y^u(f) \rightarrow O_x^u(f),$$

for $\gamma \in \widehat{\mathcal{M}}(x, y; -\nabla f)$. Hence

$$\rho_{\gamma} \otimes \tau_{\gamma}^{\vee} : O_x^M \otimes O_x^u(f)^{\vee} \rightarrow O_y^M \otimes O_y^u(f)^{\vee}.$$

Thus

$$\delta : MC^i(f; O^M) \rightarrow MC^{i+1}(f; O^M),$$

is given on $O_x^M \otimes O_x^u(f)^{\vee}$ by

$$\delta = \sum_y \sum_{\gamma \in \widehat{\mathcal{M}}(y, x; -f)} \rho_{\gamma} \otimes \tau_{\gamma}^{\vee}.$$

Poincare duality. There is a commuting square

$$\begin{array}{ccc} O_x^u(-f) & \xrightarrow{\tau_\gamma} & O_y^u(-f) \\ \downarrow & & \downarrow \\ O_x^M \otimes O_x^u(f)^\vee & \xrightarrow{\rho \otimes \tau_\gamma^\vee} & O_y^M \otimes O_y^u(f)^\vee. \end{array}$$

Consequently,

$$MH_i(-f; \mathbb{Z}) \cong MH^{m-i}(f; O^M).$$

9 Spectral Sequence

Let (C_*, ∂) be a chain complex with a filtration

$$0 = F_1 C_* \subset F_0 C_* \subset F_{-1} C_* \subset \cdots \subset F_n C_* = C_*,$$

so that

$$\partial(F_p C_k) \subset F_p C_{k-1}.$$

Define

$$E_{pq}^0 = F_p C_{p+q} / F_{p+1} C_{p+q}.$$

Then

$$0 \rightarrow F_{p+1} C_* \rightarrow F_p C_* \rightarrow F_p C_* / F_{p+1} C_* \rightarrow 0$$

gives a differential d^0 and hence

$$E_{pq}^1 = H_{p+q}(F_p C_* / F_{p+1} C_*).$$

The next differential is

$$d^1 = \pi^1 \circ \delta^1 : E_{pq}^1 \rightarrow E_{p-1, q}^1.$$

Proceeding inductively gives a spectral sequence (E^r, d^r) associated to the filtration. The red arrows in the notes describe the maps δ^0 , δ^1 , and π^1 in the exact couples.

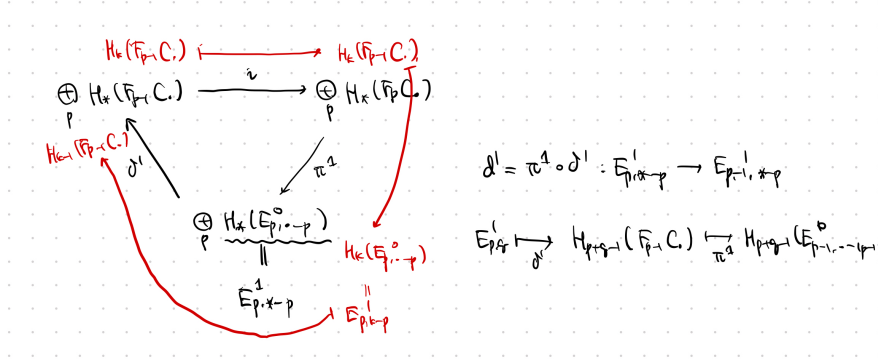


Figure 16: Spectral Sequence

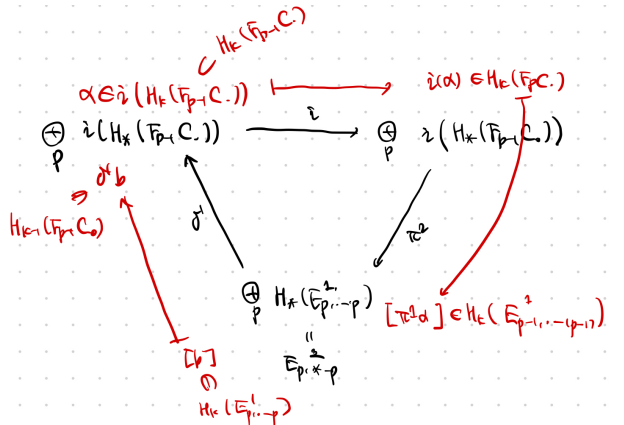


Figure 17: Spectral Sequence

To check that

$$d^r b \in \text{im } i(H_*(F_{p-2}C_*)),$$

one verifies the relations

$$i \circ j = 0, \quad \pi^2 \circ i = 0, \quad j \circ \pi^2 = 0.$$

Then

$$d^r b \in \text{im } i \iff j^r b \in \ker \pi^2 \iff j^r b = 0.$$

If $b \in H_k(F_p C_*)$ is a cycle, the corresponding higher differential has target

$$E_{p-r, q+r-1}^r.$$

The picture of lattice points indicates the bidegrees and the shift by a differential of degree $(-r, r-1)$.

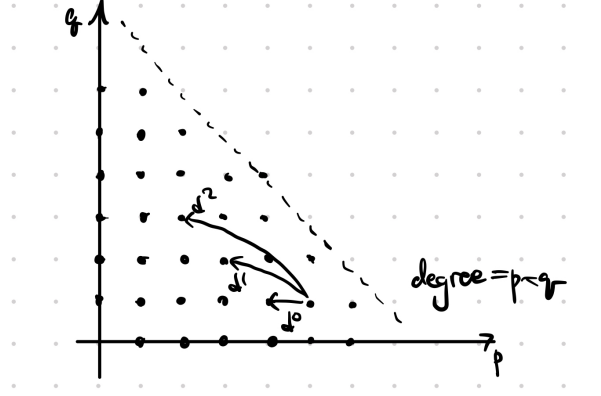


Figure 18: Spectral Sequence

If the filtration has length $n+1$, then

$$0 \subset F_0 C_* \subset \cdots \subset F_n C_* = C_*,$$

and

$$d^{n+1} = 0.$$

Thus

$$E_{pq}^{n+1} = E_{pq}^\infty.$$

There is a filtration on homology

$$0 \rightarrow H_*(F_0 C_*) \rightarrow H_*(F_1 C_*) \rightarrow \cdots \rightarrow H_*(C_*),$$

and

$$F_p H_*(C_*) := \{\alpha \in H_*(C_*) : \exists \tilde{\alpha} \in H_*(F_p C_*), [\alpha] = \tilde{\alpha}\}.$$

Then

$$E_{pq}^\infty \cong \frac{F_p H_k(C_*)}{F_{p-1} H_k(C_*)},$$

which is written in the notes as

$$E_{pq}^\infty \Rightarrow H_*(C_*).$$

10 Morse-Bott Theory

A function

$$h : M \rightarrow \mathbb{R}$$

is Morse-Bott if

1. the critical set decomposes as

$$\text{Crit } h = \bigsqcup_i C_i,$$

where each C_i is closed,

2. for every $p \in C_i$,

$$T_p C_i = \ker \text{Hess}_p(C_i).$$

Let

$$f|_{\text{Crit } h} \rightarrow \mathbb{R}$$

be Morse. For $p \in \text{Crit } f$ define

$$\text{ind}_{(h,f)} p = \text{ind}_h p + \text{ind}_f p.$$

Then

$$MBC_i(h, f) = \bigoplus_{\substack{p \in \text{Crit } f \\ \text{ind}_{(h,f)} p = i}} \mathbb{Z}_2 \langle p \rangle,$$

with a differential

$$\partial : MBC_i(h, f) \rightarrow MBC_{i-1}(h, f)$$

defined by counting cascades.

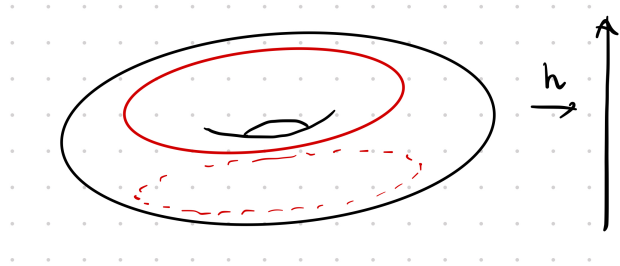


Figure 19: Morse Bott function

The Morse-Bott homology is

$$MBH_i(h, f) = H_i(MBC(h, f), \partial).$$

One has

$$MBH_i(h, f) \cong MH_i(M, f).$$

Breaking. The drawings distinguish true breaking from non-breaking cascades.

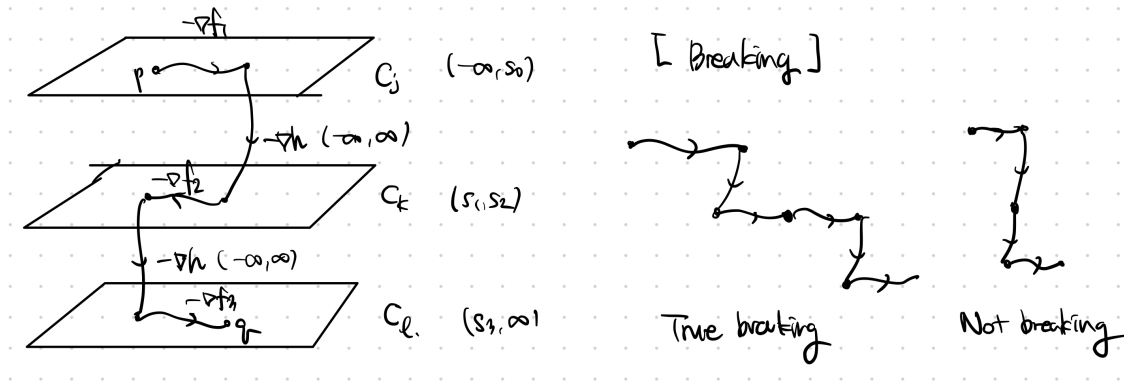


Figure 20: Breaking on Morse Bott

10.1 Leray-Serre Spectral Sequence

Let

$$\begin{array}{ccc} F & \longrightarrow & E \\ & & \downarrow \pi \\ & & B \end{array}$$

be a fiber bundle.

Theorem 29. Let F, E, B be closed manifolds and assume

$$\pi_1(B) = 0.$$

Then there exists a spectral sequence (E_{pq}^r, d^r) such that

$$E_{pq}^2 \cong H_p(B) \otimes H_q(F),$$

and it converges to

$$E_{pq}^r \Rightarrow H_*(E).$$

Let $h : B \rightarrow \mathbb{R}$ be Morse and let

$$\text{Crit } h = \bigsqcup_i \{b_i\}.$$

Then

$$h \circ \pi : E \rightarrow \mathbb{R}$$

is Morse-Bott, with

$$\text{Crit}(h \circ \pi) = E_{b_1} \sqcup \cdots \sqcup E_{b_k}.$$

Choose $f_E : E \rightarrow \mathbb{R}$ Morse so that $f|_{E_{b_i}}$ is Morse, and choose generic metrics g_B on B and g_E on E . Using a connection, one has

$$T_e E = V_e \oplus H_e \cong E_b \oplus T_b B \cong F \oplus T_b B,$$

so

$$g_E = g_F \oplus g_B.$$

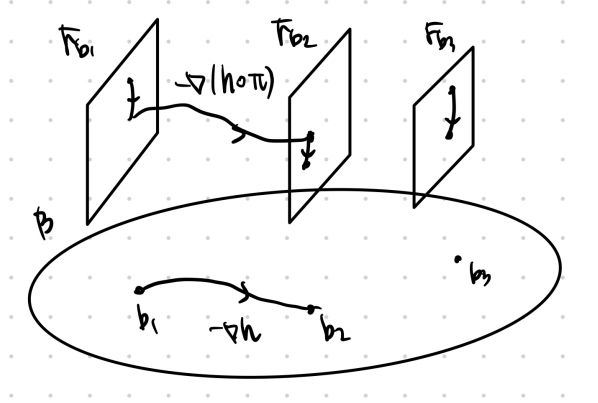


Figure 21: On the proof of Leray-Serre Spectral Sequence

Choose the filtration

$$F_p C_k = \bigoplus_{\substack{x \in \text{Crit } f \\ \text{ind}_h(x) \leq p}} \mathbb{Z}_2 \langle x \rangle.$$

Then

$$\partial(F_p C_k) \subset F_p C_{k-1}$$

because $\text{ind } h$ decreases or is preserved, never increasing. In this perspective,

$$\partial = \partial_0 + \partial_1 + \partial_2 + \cdots,$$

where ∂_i decreases $\text{ind } h$ by i . Thus

$$0 = \partial^2 = \partial_0^2 + (\partial_0 \partial_1 + \partial_1 \partial_0) + \cdots.$$

Remark 4. If $E \cong F \times B$, then

$$\partial = \partial_0 + \partial_1.$$

Now calculate the spectral sequence:

$$E_{pq}^0 = F_p C_{p+q} / F_{p-1} C_{p+q} \cong MC_p(h) \otimes MC_q(f_F).$$

Here $d^0 = \partial_0 = id \otimes \partial_{f_F}$, so

$$E_{pq}^1 = H_{p+q}(F_p C_* / F_{p-1} C_*) = MC_p(h) \otimes MH_q(f_F)$$

in field coefficients. Therefore

$$E_{pq}^1 \cong MC_p(h) \otimes H_q(F).$$

The d^1 map comes from the horizontal Morse differential. Intuitively,

$$F_p C_{p+q} \cong \frac{F_p C_{p+q}}{F_{p-1} C_{p+q}} \oplus F_{p-1} C_{p+q},$$

and the differential has the block form

$$\begin{pmatrix} \partial_0 & 0 \\ \Delta & * \end{pmatrix},$$

where Δ represents $\partial_1 + \partial_2 + \dots$. The diagrams in the notes track the components

$$(\alpha, 0), \quad \Delta\alpha, \quad \partial_0 \Delta\alpha = 0.$$

After passing to homology, this yields the map

$$H_{p+q}(F_p C_* / F_{p-1} C_*) \xrightarrow{\delta^1} H_{p+q-1}(F_{p-1} C_*).$$

The red arrows indicate the image of α , $\Delta\alpha$, and the class $[\Delta\alpha] = [d^1\alpha]$.

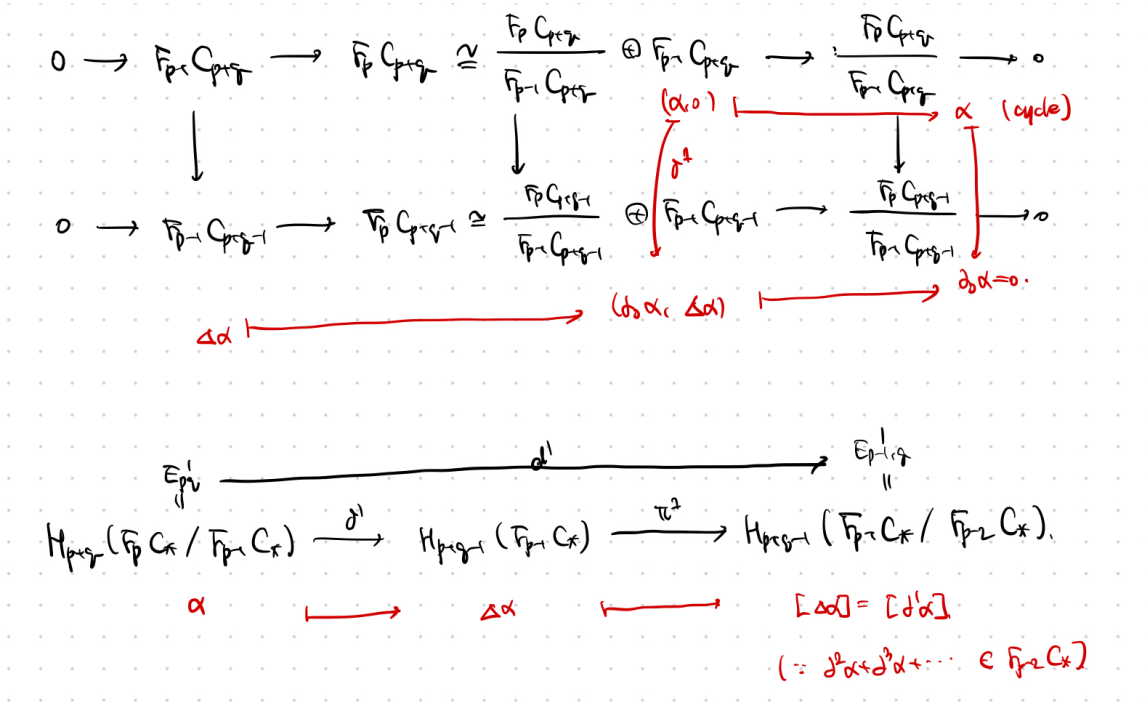


Figure 22: Spectral Sequence

Claim.

$$(E^2, d^2) \cong (MC_*(h) \otimes H_*(F), \partial_h \otimes id).$$

Sketch. Let

$$u : \mathbb{R} \rightarrow B$$

be a negative gradient trajectory

$$\partial_s u = -\nabla h(u), \quad u(\pm\infty) = b_{\pm},$$

with

$$\text{ind}_h b_- - \text{ind}_h b_+ = 1.$$

Choose a neighborhood U and trivialize

$$E|_U \cong F \times U.$$

Since $\pi_1(U) = 0$, the connection is trivial on U , so

$$TE|_U \cong H \oplus V \cong TU \oplus TF.$$

This is possible since $\pi_1(U) = 0$.

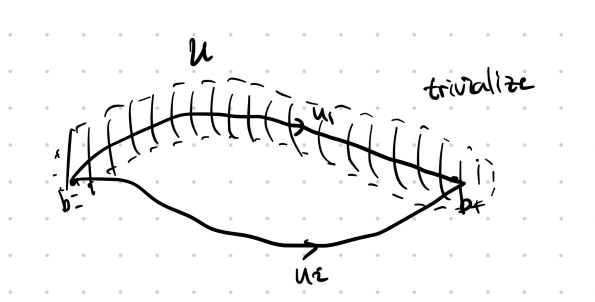


Figure 23: Trivializing

Lifting u to $\tilde{u}$ in E , the lift takes values in

$$\tilde{U} \subset U \times \mathbb{R}^q.$$

Projection to F gives the same point because we have trivialized U . Hence $-\nabla(h \circ \pi)$ moves in the same point of the fiber.

If $x_{\pm} \in \text{Crit } f$ with

$$1 - \text{ind}_{(h,f)} x_- - \text{ind}_{(h,f)} x_+ = 0,$$

then

$$\text{ind}_h b_- - \text{ind}_h b_+ = 1,$$

so the fiber index is unchanged. Therefore

$$E_{pq}^2 \cong MH_p(h) \otimes H_q(F) \cong H_p(B) \otimes H_q(F).$$

□

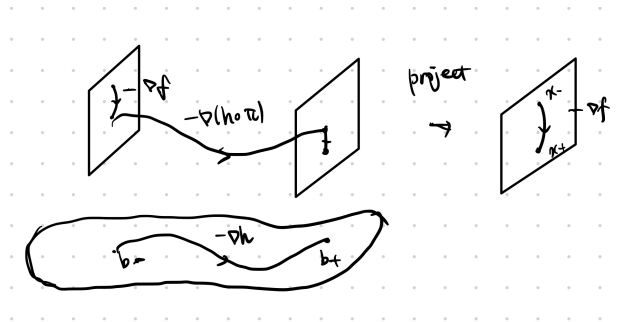


Figure 24: On the proof of Leray-Serre Spectral Sequence

11 Persistence Modules and Barcodes

A persistence module (V, π) consists of

$$V = \{V_t\},$$

where each V_t is a finite-dimensional vector space over a field $\mathbb{F}$, together with linear maps

$$\pi_{s,t} : V_s \rightarrow V_t, \quad s \leq t,$$

satisfying

1. $\pi_{s,t} = \pi_{r,t} \circ \pi_{s,r}$,
2. for finite points in $\mathbb{R}$, there is a neighborhood U such that $\pi_{s,t}$ is an isomorphism for $s \leq t$ in U ,
3. if $s \leq t$ are sufficiently close, then $\pi_{s,t}$ is an isomorphism,
4. there is s_- such that $V_s = 0$ for $s < s_-$.

Then

$$\{P_{s,t} = \text{im } \pi_{s,t}\}$$

is called the persistent homology of (V, π) .

A morphism

$$A : (V, \pi) \rightarrow (W, \theta)$$

is a family of maps $A_s : V_s \rightarrow W_s$ such that the squares commute:

$$\begin{array}{ccc} V_s & \xrightarrow{\pi_{s,t}} & V_t \\ \downarrow A_s & & \downarrow A_t \\ W_s & \xrightarrow{\theta_{s,t}} & W_t \end{array}$$

A barcode is a finite multiset of intervals

$$\mathcal{B} = \{I_1, \dots, I_k\}.$$

Theorem 30 (Normal form). *For any persistence module (V, π) , there exists a barcode $\mathcal{B}$ such that*

$$V \cong \bigoplus_i \mathbb{F}(I_i).$$

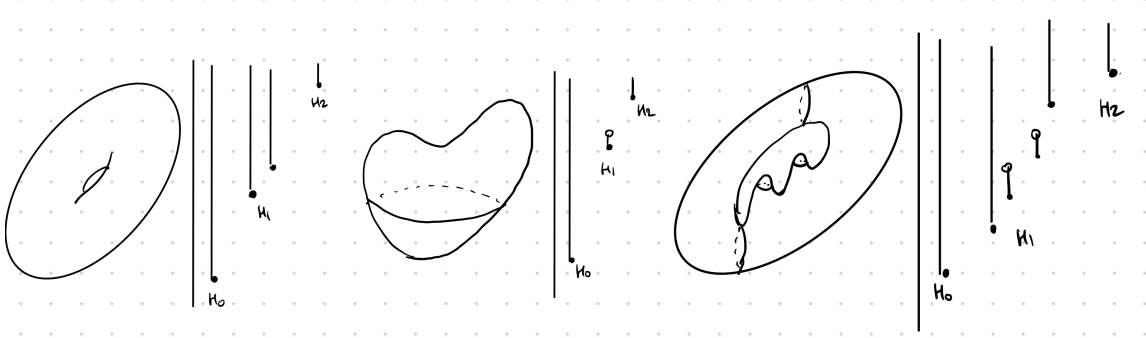


Figure 25: Examples of Persistent Barcodes

For a Morse function f , one can consider

$$V(t) = \{MH^{(-\infty, t)}(f)\}_{t \in \mathbb{R}}.$$

Then shifting gives

$$(V[\delta])_t = V_{t+\delta}, \quad \pi[\delta]_{s,t} = \pi_{s+\delta, t+\delta}.$$

Two persistence modules (V, π) and (W, θ) are interleaved if there are morphisms

$$F : V \rightarrow W[\delta], \quad G : W \rightarrow V[\delta]$$

such that

$$G[\delta] \circ F = \text{Sh}^{2\delta}, \quad F[\delta] \circ G = \text{Sh}^{2\delta}.$$

The interleaving distance is

$$d_{\text{int}}(V, W) = \inf\{\delta > 0 : V, W \text{ are } \delta\text{-interleaved}\}.$$

11.1 Stability and Barcodes

Proposition 4. For Morse functions f, g on M ,

$$d_{\text{int}}(V(f), V(g)) \leq \inf_{\varphi \in \text{Diff}(M)} \|f - \varphi^* g\|_{C^0}.$$

Proof. If φ is a diffeomorphism, then

$$V(f) \cong V(\varphi^* f).$$

Thus it is enough to show

$$d_{\text{int}}(V(f), V(g)) \leq \|f - g\|_{C^0}.$$

Put

$$\delta := \|f - g\|_{C^0}.$$

Then

$$f - \delta \leq g, \quad g - \delta \leq f.$$

Hence there are natural morphisms of persistence modules

$$V(g) \longrightarrow V(f)[\delta], \quad V(f) \longrightarrow V(g)[\delta].$$

The composites are the corresponding 2δ -shift maps,

$$V(f) \longrightarrow V(g)[\delta] \longrightarrow V(f)[2\delta],$$

so $V(f)$ and $V(g)$ are δ -interleaved. Therefore

$$d_{\text{int}}(V(f), V(g)) \leq \delta.$$

□

Definition 7 (δ -matching). Let B and C be barcodes. We say that B and C are δ -matched if, after throwing away all bars of length at most 2δ , there is a bijection between the remaining bars such that corresponding endpoints differ by less than δ .

The bottleneck distance is

$$d_{\text{bott}}(B, C) = \inf\{\delta > 0 \mid B \text{ and } C \text{ are } \delta\text{-matched}\}.$$

Theorem 31 (Isometry theorem). The barcode map

$$V \longmapsto \mathcal{B}(V)$$

is an isometry:

$$d_{\text{int}}(V, W) = d_{\text{bott}}(\mathcal{B}(V), \mathcal{B}(W)).$$

Question. Can a given Morse function

$$f : S^2 \rightarrow \mathbb{R}$$

be approximated in C^0 by another Morse function g with

$$\# \text{Crit}(g) = 2?$$

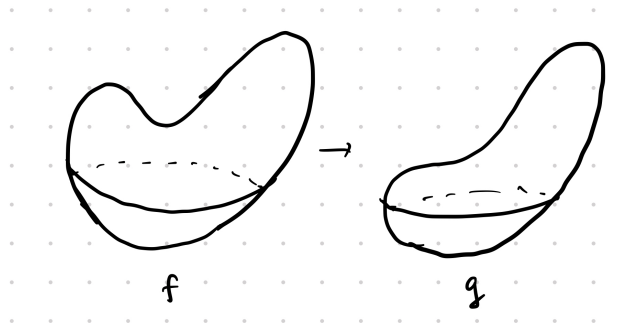


Figure 26: The problem setting

The stability and isometry theorem give

$$\|f - g\|_{C^0} \geq d_{\text{int}}(V(f), V(g)) = d_{\text{bott}}(\mathcal{B}(V(f)), \mathcal{B}(V(g))).$$

If one compares the k -th largest bar, this gives a lower bound of the form

$$d_{\text{bott}}(\mathcal{B}(V(f)), \mathcal{B}(V(g))) \geq \frac{1}{2} |\text{pr}_k(\mathcal{B}(V(f))) - \text{pr}_k(\mathcal{B}(V(g)))|.$$

In the example drawn in the notes, this lower bound is $1/2$.

Remark 5. Let $f : M \rightarrow \mathbb{R}$ be Morse. Then there exists $\varepsilon > 0$ such that, for every $g : M \rightarrow \mathbb{R}$,

$$\|f - g\|_{C^1} < \varepsilon \implies \# \text{Crit } f = \# \text{Crit } g.$$

Moreover, for sufficiently small C^2 -perturbations,

$$\|f - g\|_{C^2} < \varepsilon \implies \#\{\text{critical points of index } k \text{ for } f\} = \#\{\text{critical points of index } k \text{ for } g\}.$$

Remark 6. Let (M, g) be a closed Riemannian manifold. Consider the energy functional on the contractible loop space

$$E : \text{Contractible}(S^1, M) \rightarrow \mathbb{R}, \quad E(\gamma) = \int_{S^1} |\partial_t \gamma|_g^2 dt.$$

Its critical points are contractible geodesics. Let $\mathcal{B}(E)$ denote the barcode associated to the persistent modules

$$H_0(\{\gamma \mid E(\gamma) < a\}).$$

If there is only one infinite bar, corresponding to constant geodesics, and the lengths of the finite bars are uniformly bounded, then one obtains a uniform bound $K > 0$ for null-homotopies of null-homotopic closed geodesics. The same statement is stable for nearby metrics.

12 Geodesics Joining Two Points

Let (M, g) be a closed connected Riemannian manifold, and define

$$\Omega = \Omega(M; p, q) = \{\gamma : [0, 1] \rightarrow M \mid \gamma(0) = p, \gamma(1) = q\}.$$

The energy functional is

$$E : \Omega \rightarrow \mathbb{R}, \quad E(\gamma) = \frac{1}{2} \int_0^1 \|\dot{\gamma}\|_g^2 dt.$$

Proposition 5. A path $\gamma \in \text{Crit } E$ if and only if γ is a geodesic.

Proof. Use a variation

$$\Gamma : (-u, u) \times [0, 1] \rightarrow M$$

with

$$\Gamma(0, t) = \gamma(t), \quad \Gamma(s, 0) = p, \quad \Gamma(s, 1) = q, \quad \partial_s \Gamma|_{s=0} = X.$$

Then

$$\begin{aligned} d_\gamma E(X) &= \left. \frac{d}{ds} \right|_{s=0} E(\Gamma_s) \\ &= \int_0^1 \langle \nabla_{\partial_s} \partial_t \Gamma, \partial_t \Gamma \rangle dt \\ &= \int_0^1 \langle \nabla_{\partial_t} \partial_s \Gamma, \partial_t \Gamma \rangle dt \\ &= - \int_0^1 \langle X, \nabla_{\dot{\gamma}} \dot{\gamma} \rangle dt. \end{aligned}$$

Since X is arbitrary with $X(0) = X(1) = 0$, the critical point equation is

$$\nabla_{\dot{\gamma}} \dot{\gamma} = 0.$$

□

For the second variation, take

$$X, Y \in T_\gamma \Omega, \quad X, Y \in C^\infty(\gamma^* TM), \quad X(0) = X(1) = Y(0) = Y(1) = 0.$$

Using a two-parameter variation $\Gamma(s, r, t)$ with

$$\partial_s \Gamma|_{s=r=0} = X, \quad \partial_r \Gamma|_{s=r=0} = Y,$$

one obtains

$$\text{Hess}_\gamma E(X, Y) = - \int_0^1 g(X, \nabla_{\dot{\gamma}} \nabla_{\dot{\gamma}} Y + R(Y, \dot{\gamma}) \dot{\gamma}) dt.$$

Therefore

$$\ker \text{Hess}_\gamma E = \{\text{Jacobi fields } Y \text{ along } \gamma \mid Y(0) = Y(1) = 0\}.$$

Proposition 6. *A Jacobi field is obtained by a variation of γ through geodesics, and conversely every such variation gives a Jacobi field.*

Example: (S^2, g) . If p and q are conjugate along γ , then nontrivial Jacobi fields exist. If the geodesic is isolated, then no nontrivial Jacobi field exists.

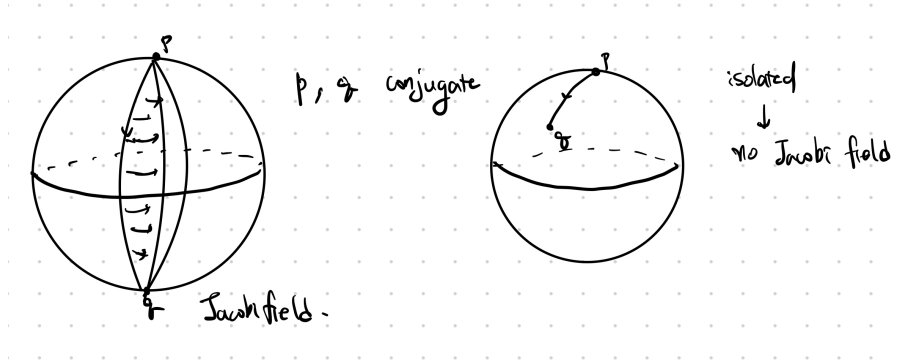


Figure 27: Existence of nontrivial Jacobi Field

Proposition 7. *For every $p \in M$, there is a dense subset*

$$M_{\text{reg}}(p) \subset M$$

such that, for every $q \in M_{\text{reg}}(p)$, the functional

$$E : \Omega(p, q; M) \rightarrow \mathbb{R}$$

has only nondegenerate critical points.

12.1 The Morse Index Theorem

Let γ be a geodesic. Define

$$\mathcal{J}_\gamma^\tau = \{\text{Jacobi fields } Y \text{ along } \gamma|_{[0, \tau]} \mid Y(0) = Y(\tau) = 0\}$$

and

$$v_\gamma^\tau = \dim \mathcal{J}_\gamma^\tau.$$

Lemma 8. *For every τ ,*

$$0 \leq v_\gamma^\tau \leq \dim M - 1.$$

Proof. The initial velocity $\dot{Y}(0)$ determines the Jacobi field Y . □

Remark 7. *For most $\tau > 0$, one has $v_\gamma^\tau = 0$. More precisely,*

$$v_\gamma^\tau > 0 \iff \gamma(0) \text{ and } \gamma(\tau) \text{ are conjugate points.}$$

Theorem 32 (Morse index theorem). *For every geodesic $\gamma \in \text{Crit } E$,*

$$\text{ind}_E(\gamma) = \sum_{\tau \in (0, 1)} v_\gamma^\tau < \infty.$$

Examples on S^2 . For a short geodesic γ_0 with no conjugate point,

$$\text{ind}_E(\gamma_0) = 0.$$

For a geodesic γ_1 passing through one conjugate point,

$$\text{ind}_E(\gamma_1) = 1.$$

For longer geodesics, one obtains further critical points

$$\text{Crit } E = \{\gamma_0, \gamma_1, \gamma_2, \dots\}$$

with indices $0, 1, 2, \dots$

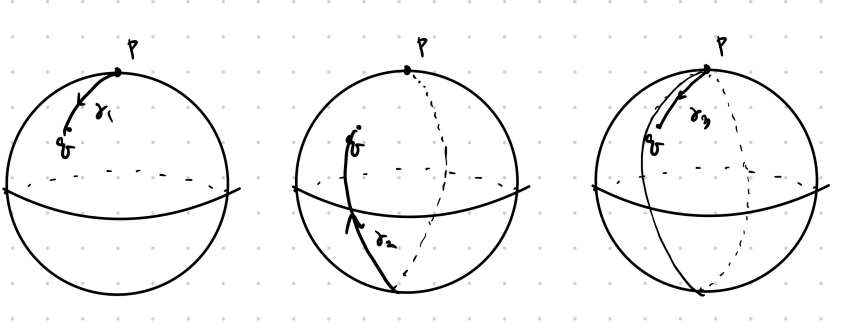


Figure 28: Morse Index Theorem

For (S^n, g) one similarly obtains critical points

$$\text{Crit } E = \{\gamma_0, \gamma_1, \gamma_2, \dots\}$$

with

$$\text{ind}(\gamma_i) = (n-1)i.$$

Remark 8. For any endpoints $p, q, r, s \in M$,

$$\Omega(M; p, q) \simeq \Omega(M; r, s).$$

Moreover,

$$\Omega^\infty(M; p, q) \simeq \Omega^0(M; p, q) \simeq \Omega^{W^{1,2}}(M; p, q),$$

by Palais' theorem.

Theorem 33. For $n \geq 2$,

$$H_i(\Omega S^n) = \begin{cases} \mathbb{Z}, & i = k(n-1), \quad k \in \mathbb{N} \cup \{0\}, \\ 0, & \text{otherwise.} \end{cases}$$

For S^1 ,

$$H_i(\Omega S^1) = \begin{cases} \mathbb{Z}^\infty, & i = 0, \\ 0, & i > 0. \end{cases}$$

Remark 9. Another way to prove the result is to use

$$\pi_k(\Omega S^n) \cong \pi_{k+1}(S^n) = \begin{cases} 0, & 1 \leq k < n-1, \\ \mathbb{Z}, & k = n-1. \end{cases}$$

Remark 10. Let $V \in C^\infty(S^n, \mathbb{R})$ be a potential function. Assume that

$$S_{V,g} : \Omega(S^n; p, q) \rightarrow \mathbb{R}$$

is Morse, where

$$S_{V,g}(\gamma) = \int_0^1 \left(\frac{1}{2} \|\dot{\gamma}\|_g^2 - V(\gamma) \right) dt.$$

Then

$$\# \text{Crit } S_{V,g} = \infty.$$

12.2 Path-Loop Fibration

Apply the Leray-Serre spectral sequence to the path-loop fibration. Let M be closed and assume

$$\pi_1(M) = 0.$$

Fix $p \in M$ and define

$$P = \{\gamma : [0, 1] \rightarrow M \mid \gamma(0) = p\}.$$

The endpoint evaluation map is

$$\text{ev} : P \rightarrow M, \quad \text{ev}(\gamma) = \gamma(1).$$

Then

$$\text{ev}^{-1}(q) = \Omega(M; p, q) \simeq \Omega(M; p, p).$$

Thus we have the path-loop fibration

$$\begin{array}{ccc} \Omega & \longrightarrow & P \\ & & \downarrow \text{ev} \\ & & M \end{array}$$

The Leray-Serre spectral sequence satisfies

$$E^r \implies H_*(P; \mathbb{Z}),$$

with

$$E_{p,q}^2 = H_p(M; \mathbb{Z}) \otimes H_q(\Omega; \mathbb{Z}).$$

In particular, for $M = S^2$, since P is contractible, the differential d^2 is nonzero and acts as an isomorphism between the relevant groups. This gives

$$\mathbb{Z} \cong H_1(\Omega S^2) \cong H_2(\Omega S^2) \cong \dots$$

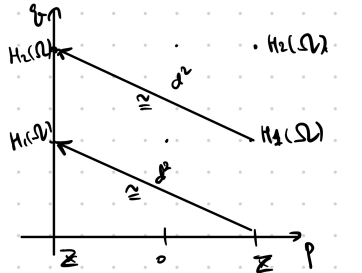


Figure 29: Spectral Sequence proof

12.3 Finite-Dimensional Approximation of the Path Space (Reference: Morse)

Let M be a complete connected Riemannian manifold. For paths

$$\omega_1, \omega_2 \in \Omega,$$

use the distance

$$\text{dist}(\omega_1, \omega_2) = \max_{t \in [0,1]} d_g(\omega_1(t), \omega_2(t)) + \left(\int_0^1 |\omega_1'(t) - \omega_2'(t)|^2 dt \right)^{1/2}.$$

Then

$$E : \Omega \rightarrow \mathbb{R}, \quad E(\omega) = \frac{1}{2} \int_0^1 |\omega'|^2 dt$$

is continuous. Put

$$\Omega^c = E^{-1}([0, c]).$$

Choose a subdivision

$$0 = t_0 < t_1 < \dots < t_k = 1$$

and define

$$\Omega(t_0, \dots, t_k) = \{\omega \in \Omega \mid \omega|_{[t_i, t_{i+1}]} \text{ is a geodesic for every } i\}.$$

Lemma 9. For a sufficiently fine subdivision of $[0, 1]$, the map

$$\text{int } \Omega^c(t_0, \dots, t_k) \rightarrow M^{k-1}, \quad \omega \mapsto (\omega(t_1), \dots, \omega(t_{k-1}))$$

is a homeomorphism onto its image.

Theorem 34. Let

$$\beta = \text{int } \Omega^c(t_0, \dots, t_k).$$

Then $E|_\beta$ is smooth. Moreover:

1. for every $a < c$, the set

$$(E|_\beta)^{-1}([0, a])$$

is a compact deformation retract of Ω^a ;

- 2.

$$\text{Crit}(E|_\beta) = \text{Crit } E \cap \text{int } \Omega^c;$$

3. for every $\gamma \in \text{Crit}(E|_\beta)$,

$$\text{ind}_{E|_\beta}(\gamma) = \text{ind}_E(\gamma).$$

Idea of Lemma. In

$$\text{int } \Omega^c(t_0, \dots, t_k),$$

the energy bound gives a uniform length bound on each subinterval. If the subdivision is sufficiently fine, then the image of each segment lies in a geodesic ball, where uniqueness of geodesics is guaranteed. Thus the path is uniquely determined by the intermediate points

$$(\omega(t_1), \dots, \omega(t_{k-1})).$$

□

The finite-dimensional approximation implies that the topology of Ω^a can be studied through

$$(E|_\beta)^{-1}([0, a]).$$

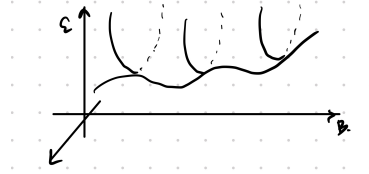


Figure 30: Interpretation of Lemma

Theorem 35. If p and q are not conjugate, then

$$\Omega(M; p, q)$$

has the homotopy type of a CW-complex.

Theorem 36. Let

$$\Omega^d = E^{-1}([0, d]).$$

Assume:

1. Ω^d is a topological submanifold;
2. every nonminimal geodesic from p to q has index at least n_0 .

Then

$$\pi_i(\Omega, \Omega^d) = 0 \quad \text{for } 0 \leq i \leq n_0 - 1.$$

Example. For S^n with p, q antipodal, the first condition is satisfied for the corresponding sublevel set.

Corollary 8. Under the assumptions of above Theorem,

$$\pi_i(\Omega^d) \cong \pi_{i+1}(M) \quad \text{for } 0 \leq i \leq n_0 - 2.$$

Corollary 9. For spheres,

$$\pi_i(S^n) \cong \pi_{i+1}(S^{n+1}) \quad \text{for } 0 \leq i \leq n - 2.$$

Proof idea of Theorem. It is enough to show that

$$\pi_i(\text{int } \Omega^c, \Omega^d) = 0$$

for sufficiently large $c > 0$. □

12.4 Bott Periodicity

Let

$$\mathfrak{u}(m) = T_I U(m) = \{X \mid X + X^* = 0\}.$$

The exponential map is

$$\exp : \mathfrak{u}(m) \rightarrow U(m).$$

Put

$$\langle A, B \rangle = \text{tr}(AB^*) \quad (A, B \in \mathfrak{u}(m)).$$

By left translation this gives a metric on $U(m)$; in fact it is bi-invariant.

Remark 11 (Observations). 1. For every $A \in T_I U(m)$ there exists $T \in U(m)$ such that

$$TAT^{-1} = \begin{pmatrix} ia_1 & & \\ & \ddots & \\ & & ia_m \end{pmatrix}, \quad a_i \in \mathbb{R}.$$

2. For every $S \in U(m)$ there exists $T \in U(m)$ such that

$$TST^{-1} = \begin{pmatrix} e^{ib_1} & & \\ & \ddots & \\ & & e^{ib_m} \end{pmatrix}, \quad b_i \in \mathbb{R}.$$

3. The exponential map

$$\exp : \mathfrak{u}(m) \rightarrow U(m)$$

is surjective.

Suppose

$$\exp(A) = -I, \quad A \in \mathfrak{u}(m).$$

After conjugating, we may assume that A is diagonal:

$$A = \text{diag}(ia_1, \dots, ia_m).$$

The condition $\exp(A) = -I$ gives

$$a_i = \pi(2k_i + 1), \quad k_i \in \mathbb{Z}.$$

Hence the length of the geodesic

$$t \mapsto \exp(tA)$$

is

$$\ell(t \mapsto \exp(tA)) = \pi \left(\sum_i (2k_i + 1)^2 \right)^{1/2}.$$

The minimal length occurs when $a_i = \pm\pi$. The space of minimal geodesics from $-I$ to I is identified with the space of subspaces of $\mathbb{C}^m$ by determining

$$\mathbb{C}^m = E_{i\pi} \oplus E_{-i\pi}.$$

For $SU(2n)$ this gives the Grassmannian

$$\text{Gr}_n(\mathbb{C}^{2n}).$$

Lemma 10 (20.2). *Every nonminimal geodesic from I to $-I$ in $SU(2n)$ has index at least $2n + 2$.*

Corollary 10. *For $0 \leq i \leq 2n$,*

$$\pi_i(\mathrm{Gr}_n(\mathbb{C}^{2n})) \cong \pi_{i+1}(SU(2n)).$$

Lemma 11 (20.4). *In the stable range there are isomorphisms*

$$\pi_i(U(m)) \cong \pi_i(U(m+1)) \cong \cdots$$

and

$$\pi_{i+1}(U(n)) \cong \pi_i(\mathrm{Gr}_n(\mathbb{C}^{2n})).$$

The preceding discussion gives Bott periodicity:

$$\pi_i(U) \cong \pi_{i+2}(U), \quad U = \lim_{n \rightarrow \infty} U(n).$$

Equivalently,

$$U \simeq \Omega^2 U,$$

where $\Omega^2 U$ is the double loop space of U .

13 Morse-Novikov Theory and Magnetic Geodesics

Motivation. Consider a force law of the form

$$F : \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}^3, \quad (q, v) \mapsto F(q, v),$$

so that an equation of motion has the form

$$F(r, \dot{r}) = \ddot{r}.$$

Lorentz force. Let

$$B : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

satisfy

$$\mathrm{div} B = 0.$$

Define

$$F_B(q, v) = v \times B(q).$$

Then

$$\langle \cdot, F_B(q, v) \rangle = \sigma_B(\cdot, v),$$

where the associated 2-form may be written as

$$\sigma_B = B_z dx \wedge dy + B_x dy \wedge dz + B_y dz \wedge dx.$$

The condition

$$d\sigma_B = 0$$

is equivalent to

$$\mathrm{div} B = 0.$$

Now let (M, g) be a closed Riemannian manifold. Given 2-form

$$\sigma \in \Omega^2(M), \quad d\sigma = 0,$$

define

$$F_\sigma : TM \rightarrow TM$$

by

$$g_q(F_\sigma(q, v), w) = \sigma_q(v, w).$$

Definition 8. *A solution of*

$$\nabla_{\dot{\gamma}} \dot{\gamma} = F_\sigma(\gamma, \dot{\gamma})$$

is called a magnetic geodesic.

If M is a surface, then g determines the area form μ_g . If

$$\sigma = f_\sigma \mu_g,$$

then

$$F_\sigma(q, v) = f_\sigma(q) i(v),$$

where i denotes rotation by $\pi/2$. Thus

$$\nabla_{\dot{\gamma}} \dot{\gamma} = -f(\gamma) i(\dot{\gamma}),$$

and the geodesic curvature is

$$k_g(\gamma) = \frac{g(\nabla_{\dot{\gamma}} \dot{\gamma}, i\dot{\gamma})}{|\dot{\gamma}|^3} = -\frac{f(\gamma)}{|\dot{\gamma}|}.$$

If $|\dot{\gamma}| = m$ is constant, then

$$k_g(\gamma) = -\frac{1}{m} f(\gamma).$$

Assume now that

$$\sigma : \pi_2(M) \rightarrow \mathbb{R}, \quad [\alpha] \mapsto \int_\alpha \sigma,$$

has discrete image $T\mathbb{Z}$ for some $T > 0$. Define the circle-valued action functional

$$S : C_{\text{Contractible}}^\infty(S^1, M) \rightarrow \mathbb{R}/T\mathbb{Z}$$

by

$$S(\gamma) = \frac{1}{2} \int |\dot{\gamma}|_g^2 dt + \int_{D^2} u^* \sigma,$$

where $u : D^2 \rightarrow M$ is a spanning disk for γ .

The critical point equation is

$$\gamma \in \text{Crit } S \iff \nabla_{\dot{\gamma}} \dot{\gamma} = F_\sigma(\gamma, \dot{\gamma}).$$

Since

$$C_{\text{Contractible}}^\infty(S^1, M) \rightarrow \mathbb{R}/T\mathbb{Z}$$

is circle-valued, one can instead study the closed 1-form dS .

We want Morse theory for maps

$$f : M \rightarrow S^1.$$

If

$$\alpha = df = f^*(d\theta),$$

then

$$\text{Crit } f = Z(df),$$

the zero set of df . The negative gradient flow equation is modified to

$$\partial_s \gamma + \nabla_g f(\gamma) = 0.$$

More generally, Morse theory can be formulated for a closed 1-form α .

A zero $p \in Z(\alpha)$ is nondegenerate if

$$\nabla \alpha : T_p M \rightarrow T_p^* M$$

is an isomorphism. Let V be the vector field defined by

$$-\alpha = g(V, \cdot).$$

For $p, q \in Z(\alpha)$ define

$$\widehat{\mathcal{M}}(p, q) = \{\gamma : \mathbb{R} \rightarrow M \mid \partial_s \gamma = V(\gamma), \gamma(-\infty) = p, \gamma(+\infty) = q\}.$$

Remark 12. *The quotient*

$$\widehat{\mathcal{M}}(p, q)/\mathbb{R}$$

need not be compact even modulo breaking, because the energy need not tend to infinity.

Proposition 8. *If*

$$\{\eta_n\} \subset \widehat{\mathcal{M}}(p, q)$$

satisfies

$$E(\eta_n) < C,$$

then compactness holds.

13.1 The Novikov Ring

Define the Novikov ring by

$$\Lambda = \left\{ \sum_{g \in \pi_1(M)} a_g g \mid a_g \in \mathbb{Z}, \forall R \in \mathbb{R} : \#\{g \mid a_g \neq 0, -\alpha(g) > R\} < \infty \right\}.$$

This allows certain infinite sums and satisfies the required flatness properties.

Choose a lift

$$\tilde{f} : \tilde{M} \rightarrow \mathbb{R}$$

such that

$$d\tilde{f} = \pi^* \alpha.$$

Then the Novikov chain group is

$$C_*^{\text{Nov}}(\alpha) = \left\{ \sum_{\tilde{p} \in \text{Crit } \tilde{f}} a_{\tilde{p}} \tilde{p} \mid a_{\tilde{p}} \in \mathbb{Z}, \forall R \in \mathbb{R} : \#\{\tilde{p} \mid a_{\tilde{p}} \neq 0, \tilde{f}(\tilde{p}) > R\} < \infty \right\}.$$

Equivalently,

$$C_*^{\text{Nov}}(\alpha) = \bigoplus_{p \in Z(\alpha)} \Lambda \cdot p,$$

a free Λ -module.

The boundary operator is defined by counting lifted flow lines:

$$\partial^{\text{Nov}} : C_*^{\text{Nov}}(\alpha) \rightarrow C_{*-1}^{\text{Nov}}(\alpha).$$

The deck transformation action gives

$$\Lambda \times C_*^{\text{Nov}} \rightarrow C_*^{\text{Nov}}, \quad g \cdot \tilde{p} \mapsto g\tilde{p}.$$

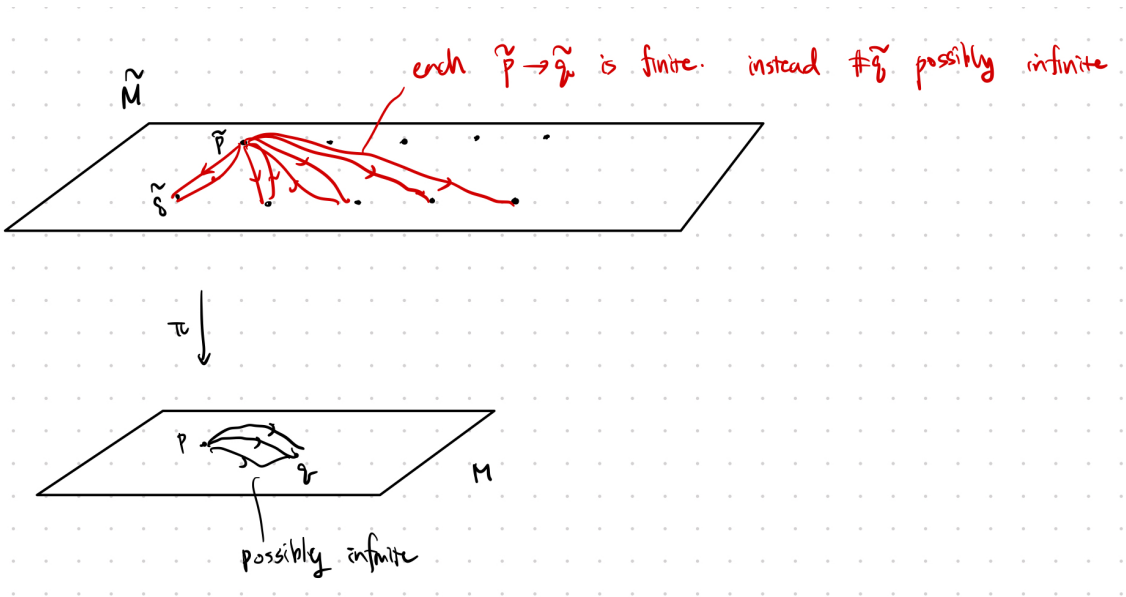


Figure 31: Morse Novikov

For each fixed lift, the number of flow lines is finite, but the number of lifts can be infinite.

This means that flow lines from p to q can be infinite in number, but after fixing a homotopy class β of paths from p to q , each count is finite. Symbolically,

$$\sum_{\beta} n_{\beta}(p \rightarrow q)_{\beta}$$

appears in the Novikov differential.

One has the algebraic description

$$H_*^{\text{Nov}}(\alpha) \cong H_*(C^{\text{cell}}(\tilde{X}) \otimes_{\mathbb{Z}[\pi_1(X)]} \Lambda).$$

Example: $M = S^1$. First consider a closed 1-form α with no zeros. Then

$$\Lambda = \mathbb{Z}((t)) = \left\{ \sum_{n=N}^{\infty} a_n t^n \mid N \in \mathbb{Z} \right\},$$

the Laurent series ring. Since

$$Z(\alpha) = \emptyset,$$

we get

$$H_*^{\text{Nov}}(\alpha) = 0.$$

Now consider

$$\alpha = df, \quad f : S^1 \rightarrow S^1.$$

Let $p, q \in Z(\alpha)$ be the two critical points indicated in the diagram.

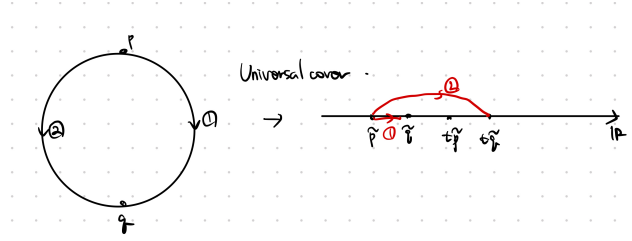


Figure 32: Example

On the universal cover the lifted critical points satisfy

$$\partial^{\text{Nov}} \tilde{p} = \tilde{q} - t\tilde{q} = (1 - t)\tilde{q}.$$

Since $(1 - t)$ is invertible in the Novikov ring, for example with inverse represented by a Laurent series, the class of $\tilde{q}$ is a boundary. Hence

$$H_*^{\text{Nov}}(\alpha) = 0.$$

Finally, if

$$f : S^1 \rightarrow \mathbb{R}$$

is real-valued, then

$$\Lambda = \mathbb{Z}[t, t^{-1}].$$

In this exact case the Novikov homology reduces to the usual Morse homology with coefficients in the group ring; in particular, the notes record

$$H_*^{\text{Nov}}(df) \cong \Lambda\langle q \rangle.$$